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Paleomagnetic study of late Miocene through Pleistocene igneous rocks from the southwestern USA: Results from the historic collections of the U.S. Geological Survey Menlo Park laboratory

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[1] Seventy sites from the southwestern United States provide paleomagnetic results that meet certain minimum criteria and can be considered for the Time-Averaged Field Initiative (TAFI). The virtual geomagnetic poles for these 70 units are circularly distributed, and their mean is nearly coincident with the rotational axis. When other published data for the southwestern United States are included (N = 146), the virtual geomagnetic poles are again circularly distributed, but their mean is significantly displaced from the rotational axis. Whichever of these data sets is used, the mean poles for normal- and reversed-polarity data differ by ~170° and are not antipodal at greater than 95% confidence. When the data are separated into specific age groups, the 95% confidence limits about the mean poles for the Brunhes, Matuyama, combined Gauss/Gilbert, and late Miocene intervals all include the rotational axis. Angular dispersion about these four mean poles increases systematically with increasing age and is consistent with paleosecular variation Model "G."

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1. Introduction

[2] Two major efforts in paleomagnetic research at the U.S. Geological Survey (USGS) during the 1960s were directed toward (1) the development of a K-Ar polarity timescale [e.g., *Glen*, 1982] and (2) a better understanding of the paleosecular variation of the geomagnetic field, particularly its dependence upon latitude. Volcanic rocks from many localities worldwide were sampled during this period. Often only the magnetic polarity and K-Ar age of the rocks sampled were published [*Cox and Dalrymple*, 1967; *Cox et al.*, 1963a, 1963b; *Dalrymple*, 1963; *Dalrymple and Hirooka*, 1965; *Dalrymple et al.*, 1965, 1967; *Doell et al.*, 1966]. In other instances, the rocks sampled proved to be unsuitable for age determinations and thus were not used in establishing the polarity time-

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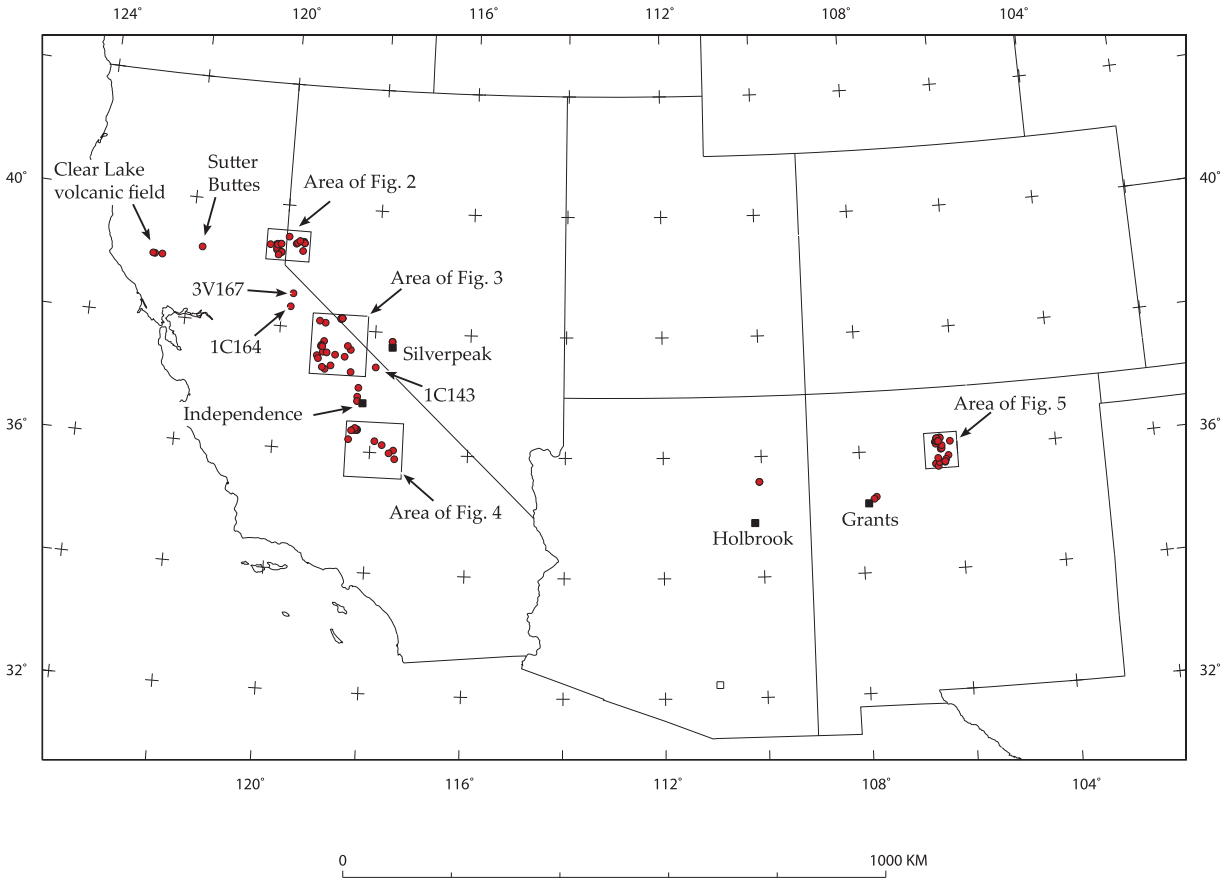


Figure 1. Index map showing the region covered by the current study. Red dots are site locations given in Table 1 and described in Appendix A.

scale. Consequently, many of these magnetic data were not published although most were included in the mean virtual geomagnetic poles (VGPs) shown on diagrams in the geomagnetic secular variation studies of *Doell* [1970, 1972a], *Doell and Dalrymple* [1973], and *Doell and Cox* [1971, 1972]. All previously unpublished data for the southwestern United States are here reevaluated for conformity to modern standards; additional measurements and analyses have been performed where necessary, and results are reported herein.

2. Methods

[3] Paleomagnetic samples from 96 localities (Figure 1) were obtained using a gasoline-powered portable core drill. Each site consisted of a single lava flow or cooling unit, typically with 8 to 10 independently oriented cores spread out over some tens of meters of outcrop. Cores were oriented with a magnetic compass and corrections applied to the measured azimuths were obtained by back sighting to one or more distant landmarks. The natural remanent magnetization (NRM) of one specimen from each core was measured using a spinner magnetometer [*Doell and Cox*, 1965]. Because the most common anomalous components of magnetization in young volcanic rocks of the western United States are the result of lightning strikes, alternating-field (AF) demagnetization was the preferred technique of magnetically cleaning the samples. Progressive demagnetization experiments were performed on one or more representative specimens from each lava flow using either three-axis tumbling demagnetizers [*Doell and Cox*, 1967], or their replacement, four-axis instruments. Because it was recognized that both types of instruments could impart a spurious component of magnetization along their innermost tumbler axes, samples commonly were demagnetized twice at the highest increments of alternating field used. For the second of these demagnetization pairs, the z axis of the sample was reversed 180° with respect to the innermost tumbler axis. This double-demagnetization technique was used multiple times dur-

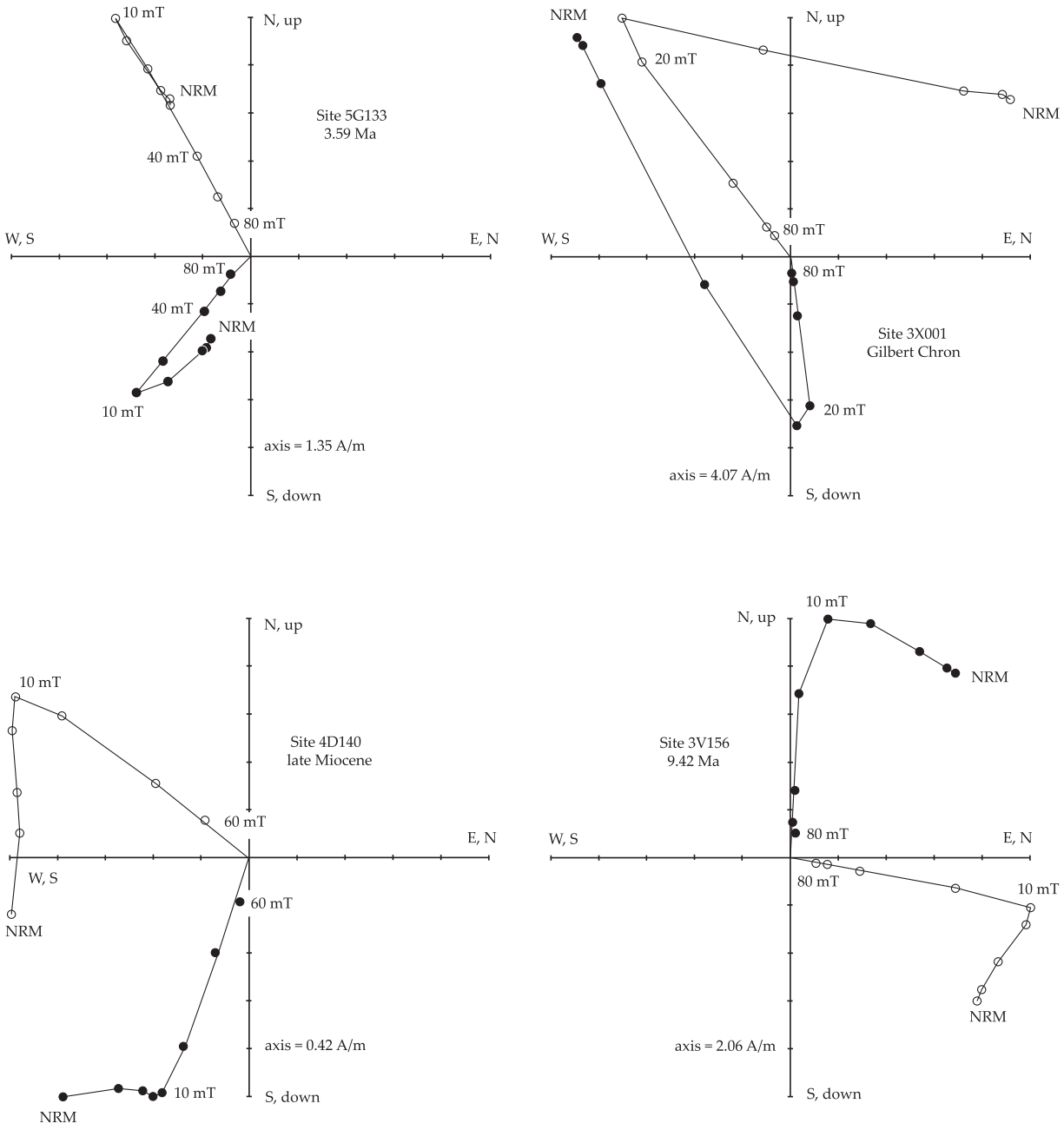


Figure 2. Orthogonal projection of remanence vector endpoints during alternating-field demagnetization of representative samples from the western United States. Open (solid) circles are projections into the vertical (horizontal) plane.

ing an experiment if any unsystematic behavior that could indicate low magnetic stability was noted. Any large discrepancy between demagnetization pairs led to rejection of the data although, in subsequent reevaluation, some were accepted after averaging using the method of *Hillhouse* [1977]. The instruments and procedures used were effective in isolating the characteristic magnetization directions in even the oldest sites sampled as shown in the representative vector component diagrams in Figure 2. Peak alternating fields to magnetically clean the sites were then chosen on the basis of behavior during the progressive demagnetization experiments (the stable endpoint method).

[4] If the NRM directions from a single lava flow were tightly grouped and progressive AF demagnetization did not exhibit appreciable changes, the normal practice in the Menlo Park laboratory was to avoid subjecting the unit to further magnetic cleaning. The main purpose was to preserve the samples for further experiments if desired. In today's world, however, such data did not meet the minimum requirements for inclusion in many of the recent data compilations (see the arguments presented by *McElhinny and McFadden* [1997]), although subsequent work shows that all NRM data should not automatically be discarded. *Tauxe et al.* [2004], for example, subjected archived core samples from 36 lava flows from Antarctica [*Mankinen and Cox*, 1988] to both AF and thermal demagnetization. Calculating a mean VGP from the original data for those same 36 sites [*Mankinen and Cox*, 1988] yields a 0.25° difference in mean VGP between "old" and "modern" methods. The Honolulu Volcanic Series (Oahu, Hawaii), originally studied by *Doell* [1972b], was resampled by *Herrero-Bervera and Valet* [2002] in order to apply modern methods to the analysis of those rocks. A new mean VGP calculated from their data differs from the original by 2.5° and is not significant at the 95% confidence level. Similar results have been reported for the Aleutian Islands [*Stone and Layer*, 2006] and Nunivak Island [*Coe et al.*, 2000], Alaska. The VGP for a new collection from the Newer Volcanics of Victoria, Australia [*Opdyke and Musgrave*, 2004] is within 4° of the VGP determined for NRM data from the original collection [*Irving and Green*, 1957] and within 6° of a collection cleaned using the stable endpoint method [*Aziz-ur-Rahman*, 1971]. For a direct comparison of results obtained using the stable endpoint method and the now more commonly used principal component analysis [*Kirschvink*, 1980], see *Beck et al.* [2001]. Nevertheless, all data reported herein have been subjected to some level of magnetic cleaning in order to more closely conform to modern standards.

3. Age Constraints

[5] Most of the geologic and chronologic information relevant to the sampled localities was published years after the paleomagnetic sampling was completed. Cox, Doell, and Dalrymple relied partly upon guidance by other USGS geologists about where rocks suitable for their studies might be found. After sampling, reliable sites were then dated by Dalrymple in the potassium-argon dating facility that had just been established within the paleomagnetic laboratory [*Glen*, 1982]. Radiometric ages that were published are given with their appropriate references in Appendix A. Sites that have not been dated, or for which the available determinations were inconclusive, are assigned to a "most-likely" polarity chron using all available geologic and age information as described below.

3.1. California-Nevada

3.1.1. Tahoe-Truckee Area

[6] The lava flows northwest of Lake Tahoe (Figure 3) postdate major uplift of the Sierra Nevada and appear to predate the oldest glaciation. Relative ages were determined by *Birkeland* [1963], who considered the basalt flows at Alder Hill to be among the oldest. One of these, the flow at site 2D001 (normal polarity) was originally reported by *Birkeland* [1963] to be 2.3 Ma old, but it was later found that the dated sample came from a nearby reversed polarity flow [*Cox et al.*, 1964]. However, the Alder Hill flow probably belongs to the Gauss Normal Chron because *Birkeland* [1963] notes that all of the younger flows of the area have reversed polarity. The only other normal polarity flow sampled in the area was the lower Watson Creek flow (site 3D107) dated at 2.5 Ma. The dated reversed-polarity volcanics range in age from the upper Watson Creek flow (site 3D101) at 2.2 Ma, to the Hirshdale (site 3D138) and Bald Mountain (site 2D072) latites at about 1.3 Ma. Site 2D005 is a second locality within the Hirshdale olivine latite. The Big Chief basalt (site 2D066) is probably younger than the basal Tahoe City latite (site 3D115; 1.9 Ma) but older than some of the other flows of that unit. The Bald Mountain olivine latite (site 2D072; 1.3 Ma) probably is slightly younger than the Big Chief basalt.

3.1.2. Mount Rose–Virginia City Area

[7] The basalt and basaltic andesite flows of the Lousetown Formation (sites 2D010 through 2D036, Appendix A) overlie the Kate Peak Formation in this area. The Kate Peak Formation is Miocene in age, and a rhyolite flow in the upper part of the unit has yielded an age of 12.7 ± 0.2 Ma [*Silberman and McKee*, 1972]. Although units of the Lousetown Formation in California (Alder Hill, Bald Mountain, Big Chief, Hirshdale, Tahoe City, etc.) are Pliocene in age, the formation ranges in age from middle Miocene to Pliocene in Nevada. The intermediate-polarity basalt flow at site 2D025

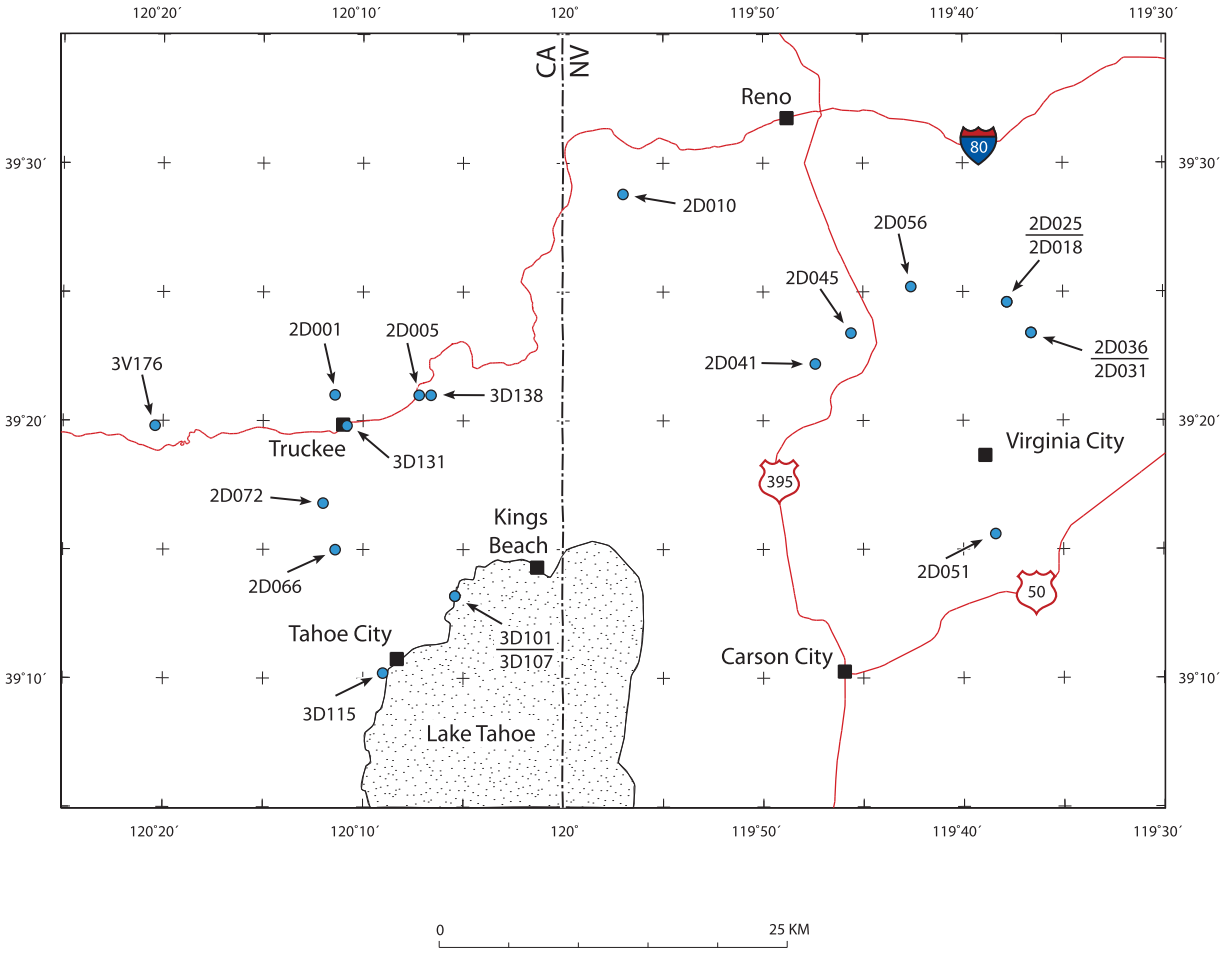


Figure 3. Index map to the Lake Tahoe–Virginia City region. Colored dots are site locations with identifiers as given in Table 1 and Appendix A.

(Figure 3), overlying the reversed-polarity site 2D018, has yielded an age of 7.08 ± 0.19 Ma [*Dalrymple et al.*, 1967]. Thus all the basalt flows sampled here are probably Miocene in age.

[8] Basaltic andesite flows in the eastern part of the Steamboat Hills were included within the Lousetown Formation by *White et al.* [1964], who thought that they were intermediate in age between flows at the type locality and the Pleistocene basalts of McClellan Peak, one of which (site 2D051) has yielded an age of 1.16 ± 0.04 Ma [*Doell et al.*, 1966]. One of these andesite flows in the Steamboat Hills, approximately 2 km from undated site 2D045, has yielded ages of about 2.5 Ma [*Silberman et al.*, 1979]. Reversed-polarity site 2D045 probably erupted during the early Matuyama Chron.

[9] Intermediate-polarity site 2D056 was sampled on a perlite rib within a rhyolite dome that shows evidence of two distinct eruptive events. Two obsidian samples from this dome yield ages of about 3 Ma, whereas sanidine from pumiceous rhyolite at the top of the same dome has been dated at 1.16 ± 0.05 [*Silberman et al.*, 1979]. Reversed-polarity site 2D041 was sampled within the large dome at the southwestern end of Steamboat Hills that has yielded an age of 1.14 ± 0.05 [*Silberman et al.*, 1979]. Thus all sites sampled for paleomagnetic study in the Steamboat Hills area belong to the Matuyama Chron.

3.1.3. Long Valley–Mono Basin Area

[10] A number of localities were sampled in this area, some of which are shown in Figure 4. An early target in this area was the Bishop Tuff, which was important to Quaternary geologists because it was a formation that was closely associated with the earliest of the North American glaciations and because its radiometric age provided a minimum

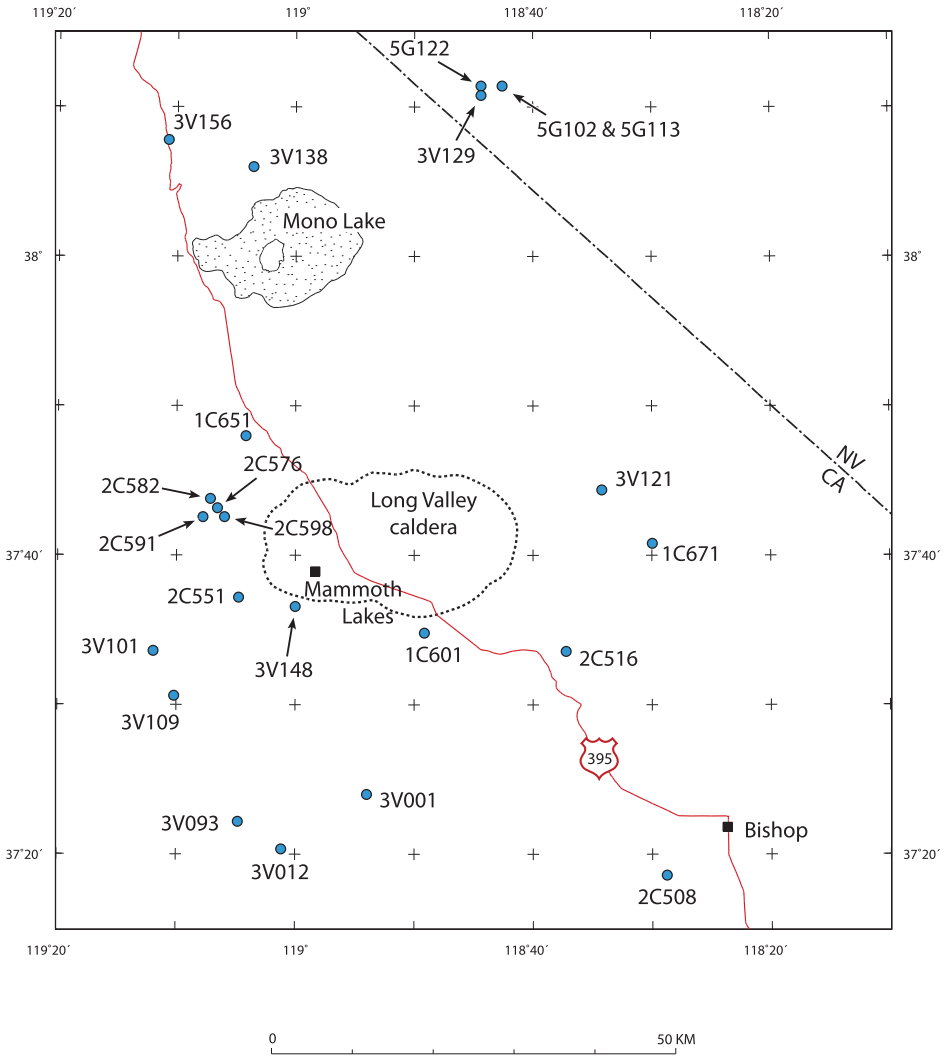


Figure 4. Index map to the Long Valley–Mono Basin region. Additional explanation as in Figure 3.

age for the Matuyama/Brunhes boundary in the early versions of the polarity timescale [*Cox et al.*, 1963a, 1963b, 1965; *Dalrymple et al.*, 1965]. Although complete paleomagnetic data were not given in those publications, the results subsequently were included in the anisotropy of magnetic susceptibility (AMS) study by *Palmer et al.* [1996] and one should refer to that publication for further details. A wide range of ages was initially obtained for the Bishop Tuff, most likely because of contamination by xenocrystic material, but the most recent determinations place its age at about 0.76 Ma using the $^{40}Ar/^{39}Ar$ method [*Izett and Obradovich*, 1994; *Sarna-Wojcicki et al.*, 2000].

[11] Most of the localities shown in Figure 4 have been dated. Site 3V148 is the basalt flow near Mammoth Mine used to define the Mammoth Reversed Polarity Subchron [*Doell et al.*, 1966]. A dated locality with a large uncertainty is normal-polarity site 2C551 within the basalt of Devils Postpile. Different plagioclase separates from the same sample have yielded ages of 0.97 ± 0.16 Ma [*Cox et al.*, 1963b; *Dalrymple*, 1964b] and 0.65 ± 0.35 Ma (G. B. Dalrymple, cited by *Huber and Rinehart* [1967]). Basalt at this site overlies the tuff of Reds Meadow [*Huber and Rinehart*, 1967], which additional dating [*Dalrymple*, 1980] and paleomagnetic analysis (site C530 [*Palmer et al.*, 1996]) have shown to be an outcrop of the Bishop Tuff. The basalt of Devils Postpile is thus Brunhes in age.

[12] Normal-polarity site 1C651 (Figure 4) was sampled in the basalt of June Lake Junction (Q_b) of *Kistler* [1966]. *Bursik and Gillespie* [1993] cite

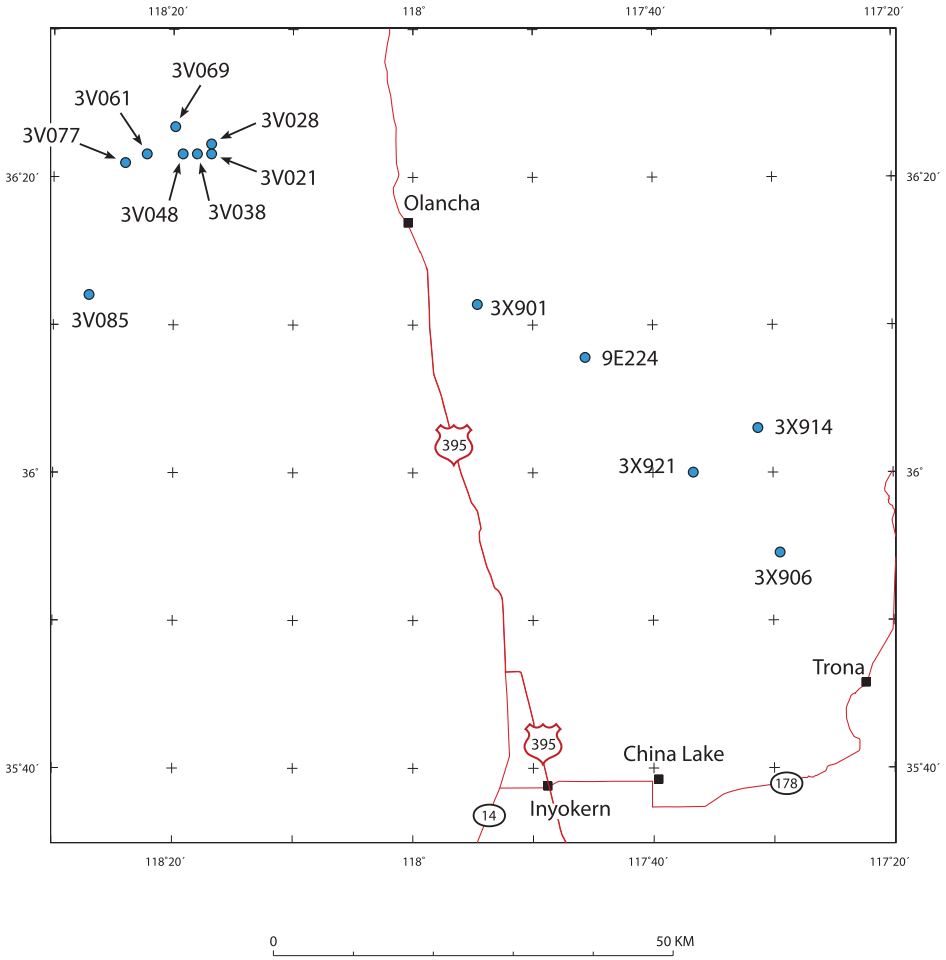


Figure 5. Index map to the Golden Trout Creek–China Lake region. Additional explanation as in Figure 3.

evidence for some basaltic eruptions at the time of Tenaya glaciation and other eruptions that postdated the Tenaya and predated the Tioga glaciation. Site 1C651 is within one of the younger basalt units and probably erupted during the short interval between the end of the Tenaya (~31 ka [*Bursik and Gillespie*, 1993]) and the beginning of the Tioga (~24.5 ka [*Benson et al.*, 1998]).

[13] Normal-polarity site 1C671 (Figure 4) was sampled in the "younger basalt (QT_{yb})" of *Rinehart and Ross* [1957]. It overlies an older alluvial unit and is in turn overlain by the Bishop Tuff and, locally, by the Sherwin Till. Without evidence to the contrary, this unit is here considered to be Brunhes in age.

[14] Three normal-polarity flows (2C576, 2C582, 2C591) were sampled (Figure 4) in a sequence of basalt flows on San Joaquin Mountain. The basalt here is overlain by rhyodacite flows forming the main mass of San Joaquin and Two Teats Mountains. Dated locality 2C598 from Two Teats Mountain stratigraphically overlies unit 2C576. This sequence of volcanic rocks in the San Joaquin–Two Teats Mountains area ranges in age from 3.2 to 2.6 Ma [*Bailey et al.*, 1976] and all units sampled belong to the Gauss Normal Polarity Chron.

[15] Site 5G122 (Figure 4) is one of four reversed polarity flows sampled near Anchorite Pass, three of which have yielded ages of 3.71 (5G113), 3.59 (5G133), and 3.43 Ma (5G102). Ages for rocks in this volcanic complex range from 3.8 to 3.2 Ma [*Gilbert et al.*, 1968], and site 5G122 can thus be assigned to the Gilbert Chron.

3.1.4. Golden Trout Creek–China Lake Area

[16] Several Quaternary basalt flows and cinder cones are present in the northern part of the area (Figure 5) in the vicinity of Golden Trout Creek.

The youngest of these emanated from Groundhog cone (site 3V048), which is postglacial and has been estimated to be between 10,000 and 5,000 years old [*Moore and Lanphere*, 1983; *Moore and Sisson*, 1985]. South Fork cone (site 3V021) erupted at 176 ± 21 ka [*Moore and Lanphere*, 1983; *Moore and Sisson*, 1985]. Products from this cone also were sampled at localities 3V038, 3V061, and 3V077. Tunnel cone (3V028) is undated but considered to be roughly contemporaneous with South Fork cone. The oldest flow sampled in this area (site 3V069; reversed polarity) is from Little Whitney cone, which erupted at 0.74 ± 0.011 Ma [*Moore and Lanphere*, 1983; *Moore and Sisson*, 1985]. Despite its radiometric age, this unit is considered to belong to the Matuyama Chron because no geomagnetic excursion occurring at about this time has been reported for the western United States [e.g., *Mankinen and Wentworth*, 2003].

[17] One normal-polarity (9E224) and four reversed-polarity (3X901, 3X906, 3X914, 3X921) lava flows were sampled toward the southern end of Figure 5, north of China Lake. The reversed-polarity lava flow at site 3X901 is within the rhyodacite southeast of Haiwee Reservoir (T_{rd}) of *Duffield and Bacon* [1981]. Ages for T_{rd} range from 2.54 Ma [*Duffield et al.*, 1980] to 2.01 Ma [*Evernden et al.*, 1964] indicating that site 3X901 dates from the early Matuyama. The reversed-polarity flow at site 3X921 belongs to the basalt of Renegade Canyon (QT_{br1}) of *Duffield and Bacon* [1981], a unit younger than QT_{br2}, which has yielded an age of 1.75 Ma [*Duffield et al.*, 1980]. Two other reversed-polarity lava flows were sampled, site 3X906 in the Argus Range northwest of Argus Peak and site 3X914 on the west side of Etcheron Valley. Neither of these units was dated, but I consider both to belong to the Matuyama (Table 1). Site 3X914 is located in an area mapped as Pleistocene volcanic rocks [*Streitz and Stinson*, 1974] although site 3X906 is within an area of undivided Cenozoic volcanic rocks [*Jennings et al.*, 1962]. The normal-polarity andesite flow 9E224 is considerably older (3.54 ± 0.08 Ma) and is part of an andesite sequence underlying a widespread ~3-Ma air fall pumice [*Duffield et al.*, 1980]. This unit is at the base of a sequence of 19 normal-polarity flows [*Pluhar et al.*, 2006] and indicates eruption near the beginning of the Gauss Chron.

3.1.5. Independence, California

[18] Two undated basalt flows were sampled along the North Fork of Oak Creek near Independence (Figure 1). Both units flowed eastward down Oak Creek Canyon postdating major uplift of the Sierra Nevada, are partially covered by moraine of the Tahoe glaciation (~46 ka, *Benson et al.* [1998]), and are considered to be Pleistocene in age [*Moore*, 1963]. Normal polarity site 1C701 and reversed polarity site 1C138 are here considered to belong to the Brunhes and Matuyama, respectively.

3.1.6. North Central California

[19] A reversed-polarity rhyolite dome of the Sutter Buttes (Figure 1) was sampled at site 2C821 (site S4 [*Cox et al.*, 1963a]). The original age determination for one of the domes at this locality (1.74 ± 0.10 Ma, sample KA65) was given by *Evernden et al.* [1957], who cautioned that the age was somewhat uncertain due to high atmospheric argon content. Later, preliminary $^{40}Ar/^{39}Ar$ determinations on sanidine from five separate rhyolite domes range in age from 1.58 to 1.52 Ma [*Swisher et al.*, 2000]. Thus the reversed-polarity for site 2C821 places its eruption during the Matuyama Chron.

[20] Two normal-polarity (2C801 and 2C804) and one reversed-polarity (2C809) sites were sampled in the Clear Lake volcanic field (Figure 1). Both normal-polarity sites are within dacite flows exposed at Horseshoe Bay and Soda Bay, respectively, which are two of the maar volcanoes along the southern shore of Clear Lake [*Hearn et al.*, 1976]. Dacite units at both sites yield ages of about 0.35 Ma [*Donnelly-Nolan et al.*, 1981]. The reversed-polarity basalt flow sampled at site 2C809 is considered Matuyama in age because all early basalt flows dated by *Donnelly-Nolan et al.* [1981] range in age from 1.96 to 1.33 Ma. The Clear Lake volcanic field is also the youngest volcanic center in the California Coast Ranges and is thought to be entirely younger than the Sonoma Volcanics (youngest published age ~2.9 Ma [*Mankinen*, 1972]) immediately to the south.

3.1.7. Silverpeak, Nevada

[21] Two superposed, normal polarity basalt flows (1C151 and 1C160) were sampled on "The Monocline," approximately 11 km north of Silverpeak (Figure 1). Rocks erupted from the Silver Peak volcanic center range in age from 6.3 to 4.9 Ma [*Robinson et al.*, 1968]. Northeast of the Silver Peak Range, these volcanic rocks overlie and interfinger with sedimentary rocks of the Esmeralda Formation. The upper part of the Esmeralda Formation is exposed at the paleomagnetic sampling locality and the basalt there overlies a biotite tuff

Table 1. Site-Average Paleomagnetic Data for Igneous Rocks From the SW United States[a]

Unit Number	N/N_o	H, mT	Incl., deg	Decl., deg	R	k	α_{95}, deg	J_{nrm}, A/m	J_{cl}, A/m	Plat., deg	Plong., deg
Brunhes Normal Polarity Chron											
1C614	8/8	10	68.1	343.6	7.9018	71	6.6	7.99	6.22	72.7	203.5
1C651	11/11	60	70.2	346.3	10.9556	225	3.1	2.08	0.87	71.0	215.7
1C671	9/10	60	47.8	305.3	8.8399	50	7.4	1.68	0.22	44.1	157.4
1C684	8/8	20	70.2	0.4	7.9717	248	3.5	2.28	1.08	72.6	242.5
1C701	6/6	10	60.6	344.9	5.9846	324	3.7	3.24	2.42	77.4	178.5
2C551	7/7	20	57.4	333.4	6.9746	236	3.9	14.0	4.52	69.1	160.1
2C801[b]	3/3	10	55.9	9.3	2.9916	238	8.0	1.33	1.25	82.2	333.7
2C804	5/5	10	56.2	0.0	4.9662	118	7.1	1.26	1.05	87.6	39.0
3V021	6/7	40	57.6	4.2	5.9939	817	2.4	1.74	0.23	86.2	301.0
3V028	9/10	80	55.4	8.8	8.9333	120	4.7	11.1	0.70	82.9	333.0
3V038	10/10	40	56.4	5.6	9.9704	304	2.8	1.89	1.02	85.5	322.4
3V048	11/13	60	53.4	359.6	10.8823	85	5.0	5.21	0.63	87.6	70.0
3V061	8/8	20	59.3	359.1	7.9866	522	2.4	1.41	1.25	86.2	230.9
3V077	7/8	20	55.3	359.7	6.9789	285	3.6	1.87	0.41	89.4	82.3
Matuyama Reversed Polarity Chron											
1C118	20/20	60	−52.0	174.9	19.5854	46	4.9	0.32	0.11	−83.9	286.5
1C138[b]	5/5	20	−25.9	148.2	4.6773	12	22.6	2.15	0.50	−53.3	241.7
2C809	6/6	10	−65.8	179.9	5.9757	306	4.7	2.53	2.40	−81.0	56.8
2C821	7/7	10	−68.4	157.1	6.9925	797	2.1	0.47	0.45	−69.8	13.8
2D005	5/5	20	−42.9	181.0	4.9664	119	7.0	1.36	0.50	−75.5	236.1
2D041[b]	4/4	10	−61.1	195.3	3.9821	167	7.1	0.03	0.03	−78.1	131.5
2D045	5/6	40	−53.2	198.8	4.9689	128	6.8	1.87	0.47	−73.9	164.9
2D051[b]	4/5	20	−46.9	177.8	3.8946	28	17.5	5.97	1.07	−78.7	250.4
2D056[b]	4/4	40	−33.1	253.1	3.9927	413	4.5	0.67	0.16	−24.2	146.3
2D066[b]	6/6	60	−51.1	187.5	5.8107	26	13.3	2.53	0.32	−80.4	198.3
2D072	5/5	20	−47.6	163.7	4.9760	167	6.0	1.61	0.70	−72.9	296.7
3D101	5/5	20	−72.1	186.9	4.9661	118	7.1	2.20	0.85	−71.6	71.9
3D115	6/6	20	−54.6	172.7	5.9542	109	6.4	1.61	0.92	−83.0	297.4
3D131	7/7	20	−53.0	160.1	6.9638	166	4.7	0.54	0.41	−73.0	316.0
3D138	6/6	10	−43.0	196.1	5.9771	218	4.5	0.96	0.94	−70.3	191.7
3V069	8/8	40	−57.1	186.0	7.9901	707	2.1	4.28	1.57	−85.0	134.6
3X209	7/7	20	−54.6	194.8	6.9496	119	5.6	10.9	5.59	−78.0	163.1
3X245	7/7	60	−52.2	195.8	6.9736	227	4.0	11.2	2.41	−76.7	172.2
3X901[b]	5/5	20	−70.8	172.3	4.9085	44	12.0	0.27	0.05	−70.3	49.0
3X906	8/8	40	−51.0	184.2	7.9775	311	3.2	8.73	3.10	−84.6	202.0
3X914	7/7	20	−45.9	183.0	6.9874	474	2.8	3.24	2.21	−80.9	225.4
3X921	7/7	20	−58.8	176.6	6.9886	525	2.6	1.61	1.41	−85.6	25.8
4D122	6/7	20	−32.6	172.3	5.9618	131	5.9	0.42	0.12	−70.4	276.0
Gauss Normal Polarity Chron											
1C601	12/12	20	59.9	20.8	11.9052	116	4.1	3.29	2.35	73.6	313.1
2C516[b]	8/8	10	81.4	98.3	7.9567	162	4.4	0.80	0.73	33.4	261.4
2C576	6/6	10	43.6	11.0	5.9687	160	5.3	1.29	1.02	74.6	20.5
2C582	9/9	20	64.6	332.4	8.7652	34	9.0	4.80	0.59	67.9	183.1
2C591	7/7	20	69.4	351.1	6.8861	53	8.4	3.05	0.24	73.5	221.8
2D001[b]	4/4	20	51.1	0.2	3.9830	176	6.9	1.54	1.17	82.4	58.7
3D107	5/5	20	57.6	26.8	4.9547	88	8.2	1.07	0.76	69.2	324.1
3V101	8/8	40	51.0	350.1	7.8600	50	7.9	5.98	0.14	80.0	117.8
3V121	8/8	20	59.0	324.9	7.9885	610	2.2	6.52	2.66	62.7	166.6
3V148	6/6	20	−47.2	183.2	5.9716	176	5.1	2.30	0.59	−80.4	223.8
3X020	8/8	20	53.4	352.4	7.9709	241	3.6	7.75	2.68	83.6	152.6
3X031	7/7	40	69.1	330.1	6.9285	84	6.6	2.42	1.17	62.7	211.1
3X038	8/8	40	78.0	341.3	7.8777	57	7.4	0.12	0.05	56.5	239.1
4D082	6/8	20	50.6	8.5	5.9762	210	4.6	4.35	2.61	81.4	15.4
4D090	7/8	20	49.9	342.1	6.9622	159	4.8	6.23	4.05	74.1	148.8
4D183	7/7	20	45.0	355.0	6.9638	166	4.7	2.46	1.22	79.8	100.1
4D191	7/8	10	45.8	352.9	6.9344	91	6.4	2.94	2.83	79.5	110.9
9E224	10/10	10	70.4	347.0	9.9837	552	2.1	1.09	0.97	69.6	220.3

Table 1. (continued)

Unit Number	N/N$_o$	H, mT	Incl., deg	Decl., deg	R	k	α_{95}, deg	J_{nrm}, A/m	J_{cl}, A/m	Plat., deg	Plong., deg
Gilbert Reversed Polarity Chron											
3V001	7/8	20	−49.0	177.8	6.9875	480	2.8	2.65	0.96	−82.3	255.6
3V012[b]	7/9	20	−46.0	163.2	6.7441	23	12.7	2.14	0.12	−72.7	300.7
3V085	8/8	20	−44.4	173.2	7.9696	230	3.7	4.72	0.91	−78.4	273.6
3V093	7/8	40	−45.0	188.2	6.9359	94	6.3	4.68	0.17	−77.1	206.0
3V109	8/8	20	−38.9	208.3	7.9302	100	5.6	20.1	1.63	−61.1	175.3
3V167	8/9	20	−56.0	169.3	7.9827	405	2.8	0.04	0.03	−81.3	320.3
3X001	8/8	40	−52.0	172.5	7.9606	178	4.2	3.92	1.17	−83.1	316.7
3X013	5/6	40	−45.4	179.5	4.9484	77	8.8	8.38	0.69	−81.3	252.7
4D174	7/8	80	65.1	357.0	6.9909	659	2.4	3.57	0.82	78.5	243.2
5G102[c]	9/9	10	−43.1	203.1	8.9063	85	5.6	18.0	18.4	−66.5	178.4
5G113[c]	5/8	20	−50.4	181.8	4.9833	239	5.0	7.47	7.43	−82.9	229.0
5G122[c]	8/8	60	−66.0	197.1	7.9718	248	3.5	2.97	0.95	−74.0	106.5
5G133	7/8	40	−58.9	212.9	6.9524	126	5.4	3.72	0.66	−64.5	137.6
Pre-Gilbert Chrons (Late Miocene, <10 Ma)											
1C143[b]	5/8	20	−60.9	209.2	4.8269	23	16.3	2.51	0.81	−67.3	131.8
1C151	5/5	20	76.9	1.9	4.9848	263	4.7	1.71	0.92	62.9	244.1
1C160[b]	3/3	10	58.0	321.9	2.9100	22	26.8	1.85	0.47	60.3	165.9
2C508[b]	5/8	40	−69.9	143.3	4.9162	48	11.2	1.44	0.38	−60.0	16.8
2D010[b]	4/5	20	38.9	45.3	3.9472	57	12.3	1.19	0.29	47.9	340.8
2D018[b]	6/7	20	−46.4	217.9	5.7651	21	14.9	0.47	0.32	−56.6	159.2
2D025[b]	6/6	40	−49.8	103.0	5.8083	26	13.4	1.65	0.07	−28.2	348.3
2D031[b]	5/5	40	−70.2	45.3	4.8710	31	14.0	2.55	0.21	−11.3	35.3
2D036[b]	5/5	20	−67.5	60.5	4.9641	111	7.3	1.70	0.38	−14.2	25.4
3V129[c]	8/8	10	−67.0	178.6	7.9645	197	4.0	3.07	2.72	−78.4	56.6
3V138[c]	8/8	20	−56.2	161.2	7.9851	470	2.6	2.64	0.61	−75.0	331.6
3V156	9/9	20	27.5	355.4	8.9666	239	3.3	4.44	2.92	66.1	71.9
3V176	7/7	20	51.5	341.1	6.9798	296	3.5	9.90	3.07	73.1	130.3
3X216	6/6	20	30.3	329.4	5.9433	88	7.2	1.62	1.16	56.4	135.9
3X252[b]	7/7	40	34.9	8.8	6.7040	20	13.7	9.99	5.34	71.5	46.7
3X259	7/7	60	55.7	349.9	6.9235	78	6.9	1.83	0.11	81.8	168.1
4D098	8/8	10	77.9	13.2	7.9881	587	2.3	4.11	3.39	58.4	263.5
4D106	8/8	40	51.9	12.1	7.9814	377	2.9	2.32	1.06	79.4	359.9
4D114	8/8	40	57.9	46.8	7.9492	138	4.7	0.42	0.13	53.1	325.3
4D130	8/8	40	71.3	348.2	7.9290	99	5.6	5.80	1.03	68.7	235.2
4D140	7/8	40	−37.5	207.4	6.8832	51	8.5	0.48	0.26	−61.9	187.8
4D148	8/8	10	−52.8	164.7	7.9521	146	4.6	0.94	0.99	−77.2	337.5
4D155	9/9	10	13.4	339.7	8.9882	678	2.0	0.82	0.83	55.6	111.1
4D165	8/8	80	56.6	1.5	7.9668	211	3.8	6.39	0.30	88.2	293.7
4D218	7/8	80	42.6	340.3	6.9629	162	4.8	3.53	0.86	69.6	134.9
4D227	8/8	40	22.8	335.3	7.9372	112	5.3	1.33	0.15	57.1	122.4

[a] *N*/*N*o, number of samples used in the analysis/total number of samples collected; *H*, peak demagnetizing field (mT); Incl., Decl., inclination, positive downward and declination, east of north, respectively, of remanence vector; *R*, resultant vector of *N* unit vectors; *k*, precision parameter $(N-1)/(N-R)$; α_{95}, 95% confidence cone about average direction; J_{nrm}, J_{cl}, geometric mean intensities before and after magnetic cleaning, respectively (A/m); Plat., Plong., north latitude and east longitude, respectively, of VGP (virtual geomagnetic pole).
[b] Sites not meeting minimum criteria (see text).
[c] Sites tectonically rotated.

with a reported age of 7.1 ± 0.3 Ma (J. F. Evernden, cited by *Robinson et al.* [1968]). The sites sampled are Miocene in age.

3.2. Arizona–New Mexico

3.2.1. Northeastern Arizona

[22] Two superposed, reversed-polarity basalt flows (sites 3X001 and 3X013), occurring between the upper and lower members of the Bidahochi Formation, were sampled approximately 75 km north of Holbrook (Figure 1). One of these (sample KA 1027) yielded a K-Ar age of 4.2 Ma [*Evernden et al.*, 1964]. Although those authors considered that to be a minimum age, both flows probably belong to the Gilbert Reversed Polarity Chron as the formation is generally considered to be Pliocene in age [*Hackman and Olson*, 1977]. Ages determined for the volcanic centers in the Holbrook

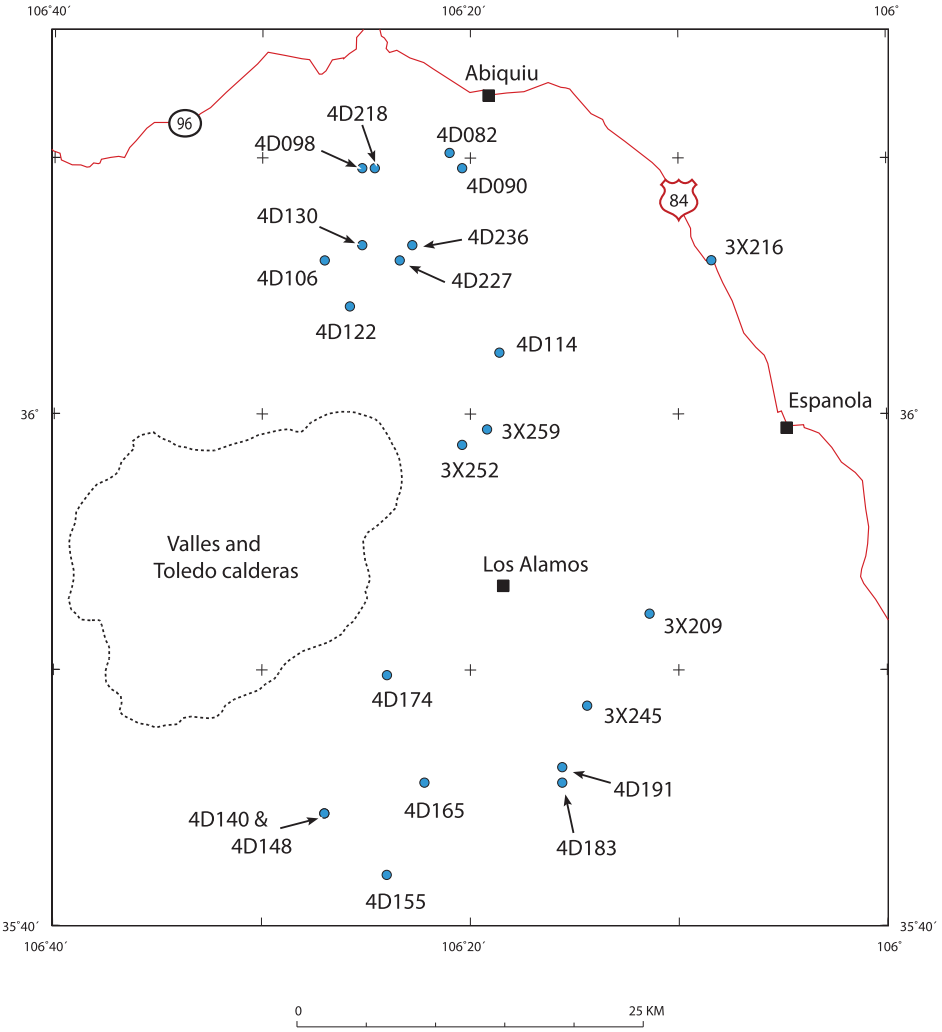


Figure 6. Index map to the Valles/Toledo calderas region. Additional explanation as in Figure 3.

region range from 2.2 to 6.87 Ma [*Luedke and Smith*, 1978].

3.2.2. Grants, New Mexico

[23] Three normal polarity sites (3X020, 3X031, and 3X038) were sampled northeast of Grants (Figure 1). The latter two were sampled near a perlite mine on the east side of Grants Mesa. Site 3X038 was dated at 2.81 ± 0.08 Ma [*Dalrymple et al.*, 1967]. Obsidian fragments in tuff underlying the lava flows on Grants Mesa and La Jara Mesa, approximately 6.2 km to the northeast have yielded ages of ~3.6 Ma [*Bassett et al.*, 1963]. Site 3X020 is from a lava flow capping La Jara Mesa. All three sites probably belong to the Gauss Normal Polarity Chron.

3.2.3. Abiquiu Area

[24] A number of localities were sampled north of Valles Caldera in the vicinity of Abiquiu (Figure 6). Normal polarity sites 4D082 and 4D090 were sampled in the unit termed "basaltic lavas of Cerros del Rio, Santa Ana Mesa, and El Alto (Q_{tb})" by *Smith et al.* [1970]. *Luedke and Smith* [1978] report an unpublished age of 2.9 ± 0.1 Ma near these two sites. *Gardner et al.* [1986] consider a minimum time span of 4.6 to 1.45 or 1.12 Ma for this unit. Both studies indicate eruption of sites 4D082 and 4D090 most likely occurred during the Gauss Normal Polarity Chron.

[25] Normal polarity sites 4D098 (7.63 ± 0.16 Ma) and 4D114 were sampled in the Lobato Basalt [*Bailey et al.*, 1969]. The age determined for site

4D098 [*Dalrymple et al.*, 1967] along with ages ranging from about 14 to 7 Ma [*Gardner et al.*, 1986; *Dethier et al.*, 1986] indicates that the Lobato Basalt is entirely Miocene in age.

[26] Undated, normal polarity site 4D218 belongs to the Tschicoma Formation [*Griggs*, 1964] as do dated sites 3X252, 3X259, 4D106, 4D130, 4D174, and 4D227, all of which also have normal magnetic polarity. This unit consists mainly of dacite flows and domes, with subordinate andesite, and ranges from about 7 to 3 Ma in age [*Bachman and Mehnert*, 1978; *Gardner et al.*, 1986; *Dethier et al.*, 1986]. The dated localities except for site 4D174 (3.77 ± 0.12 Ma) [*Dalrymple et al.*, 1967] are Miocene in age. The age for site 4D174 places it within the Gilbert Chron and perhaps representing the Cochiti Subchron. Site 4D218 is within an andesite flow, and *Gardner et al.* [1986] consider that most Tschicoma andesites may be about 7 Ma in age. Without evidence to the contrary, site 4D218 is placed among the Miocene localities in Table 1.

3.2.4. Cerros del Rio Area

[27] The localities sampled in this area are southeast of Valles Caldera (Figure 6) and belong to the "basaltic lavas of Cerros del Rio, Santa Ana Mesa, and El Alto (QT_b)." Normal polarity sites 4D183 and 4D191 most likely belong to the Gauss Chron, on the basis of the time span for the unit as described in the previous section. Reversed polarity sites 3X209 and 3X245 directly underlie the Bandelier Tuff (1.61–1.23 Ma [*Spell et al.*, 1996]) and thus could be high in the sequence and belong to the Matuyama Chron.

3.2.5. Bland District

[28] Reversed polarity sites 4D140 and 4D148 were sampled in the unit termed "Volcanic and intrusive rocks of the Bland District, undivided (T_{vi})" by *Smith et al.* [1970]. This unit includes hydrothermally altered volcanic and hypabyssal rocks of the Cochiti mining district [*Gardner et al.*, 1986]. Site 4D148 has yielded an age of 9.11 ± 0.19 Ma [*Dalrymple et al.*, 1967] and all the rocks in this unit are overlain by the volcanic rocks of the Keres Group, the youngest of which are ~6 Ma [*Gardner et al.*, 1986]. Thus both sites are Miocene in age.

3.2.6. Rio Chama Area

[29] A normal polarity basalt flow interbedded within the Santa Fe Formation [*Smith et al.*, 1970] was sampled at site 3X216 (Figure 6). The formation is Miocene in age, underlying the basalt of Chamisa Mesa (~10 Ma [*Luedke and Smith*, 1978; *Gardner et al.*, 1986]) southwest of Valles caldera, and underlying the Lobato Basalt and the basaltic lavas of Cerros del Rio toward the north and east of the caldera.

4. Results

4.1. This Study

[30] Remanent magnetization directions and associated statistical parameters for the sites listed in Appendix A are given in Table 1. A number of sites exhibit relatively large dispersions, presumably due to incomplete removal of the effects of lightning strikes. This is particularly true for samples from the Lousetown Formation (sites 2D010 through 2D036, Table 1) near Virginia City (Figure 3). Previous studies [*Heinrichs*, 1967; *Roberts and Shaw*, 1990] have shown that these lava flows have been severely affected by lightning. Although the sites in the table that have large dispersions are sufficiently accurate to indicate polarity, they clearly are not suitable for secular variation study.

[31] *Tauxe et al.* [2003] suggest that, in order to obtain the highest quality data for analysis, the minimum number of samples per site should be ≥ 5 and that the concentration factor, κ, of *Fisher* [1953] be ≥ 100 (or alternatively, $\alpha_{95} \leq 5.0°$). They also find no justification for excluding low-latitude VGPs although many researchers are using the method of *Vandamme* [1994] to calculate an optimal cutoff angle to exclude transitional data representing excursions or polarity transitions. Using $\alpha_{95} \leq 5.0°$as a criterion may be overly restrictive, however, and has the potential to bias analyses toward the youngest and least altered of sites. Analyses of very young basalt flows on Hawaii [*Doell and Cox*, 1963; *Hagstrum and Champion*, 1994] have shown that the attainable accuracy of paleomagnetic directions for any one unit was probably no more than ~3°to 5° at the 95% confidence level. For these reasons, the minimum criteria used here are $N \geq 5$ and $\alpha_{95} \leq 10.0°$ [also see *Opdyke et al.*, 2006]. VGP latitudes below 45° or 50° are commonly used as cutoff values, but values between 40° and 50° are not uncommon and could be part of normal secular variation. In this data set, site 1C671 has a VGP latitude of 44.1° (Table 1) and will be retained for further analyses. Two sites (2C516 and 2D036) have VGP latitudes that are more than 55° away from the rotational

Table 2. Statistical Analysis of Flow Average Data[a]

Data Analyzed	N	*PLA*, deg	*PLO*, deg	R	K	α_{95}, deg	S_T, deg	S_F, deg	S_u, deg	S_l, deg
Fisher [1953] statistics										
This study	70	87.4	162.9	66.4823	19.6	3.9	18.3	17.9	20.3	16.0
Normal	42	83.5	185.8	39.6222	17.2	5.5	19.5	19.2	22.6	16.7
Reversed	28	85.6	39.8	27.1321	31.1	5.0	14.5	14.0	17.2	11.8
All USGS	146	86.3	107.6	137.9398	18.0	2.8	19.1	18.6	20.2	17.2
Normal	85	86.4	156.3	80.2750	17.8	3.8	19.2	18.6	20.8	16.8
Reversed	61	83.3	73.0	57.9311	19.6	4.2	18.3	17.8	20.4	15.8
Brunhes	38	86.3	185.7	36.5000	24.7	4.8	16.3	15.6	18.5	13.5
Matuyama	30	84.5	61.0	28.7196	22.7	5.7	17.0	16.6	20.2	14.1
Gauss/Gilbert	48	85.3	82.1	45.3343	17.6	5.0	19.3	19.0	22.1	16.7
Miocene	30	82.9	129.1	27.6708	12.5	7.8	23.0	22.7	27.6	19.3

Data Analyzed	N	*PLA*, deg	*PLO*, deg	K_1	K_2	min. α_{95}, deg	max. α_{95}, deg
Bingham [*Onstott*, 1980] statistics							
This study	70	87.5	161.0	−13.2	−9.8	3.5	4.1
Normal	42	83.5	186.2	−12.6	−8.5	4.7	5.8
Reversed	28	85.5	41.6	−26.0	−12.3	3.8	5.7
All USGS	146	86.3	103.9	−12.2	−9.0	2.5	3.0
Normal	85	86.6	154.4	−11.3	−9.4	3.5	3.8
Reversed	61	83.3	71.2	−17.4	−8.4	3.3	4.8
Brunhes	38	87.1	184.5	−18.0	−11.3	4.0	5.1
Matuyama	30	84.6	58.1	−19.6	−9.9	4.3	6.2
Gauss/Gilbert	48	85.1	81.3	−13.8	−8.3	4.2	5.5
Miocene	30	82.5	131.1	−8.6	−7.0	6.9	7.8

[a] N, number of VGPs (virtual geomagnetic poles) used; *PLA*, *PLO*, north latitude and east longitude, respectively, of mean pole; R, resultant vector of N poles; k, precision parameter $(N - 1)/(N - R)$; α_{95}, 95% confidence cone about average direction; S_T, total angular dispersion about the mean; S_F, dispersion corrected for within-site dispersion [*McElhinny and Merrill*, 1975]; S_u, S_l, upper and lower 95% confidence limits, respectively, for S_F [*Cox*, 1969]; K_1, K_2, Bingham concentration parameters.

axis, truly outside the limits of normal secular variation, and these will be excluded.

[32] *King et al.* [2007] show that Eureka Valley Tuff sites 3V129 and 3V138, which are within the Excelsior-Coaldale block of the Walker Lane tectonic belt [*Wesnousky*, 2005], have undergone significant clockwise, vertical-axis rotations. Sites 5G102, 5G113, and 5G122 are in the same block (Figure 4) and have magnetization directions that are consistent with significant clockwise rotations. Results from these three sites provide new constraints on the timing of rotation of this block, indicating that the rotation occurred after 3.4 Ma. All five sites will be excluded from the calculations that follow.

[33] Of the 96 sites given in Table 1, 70 meet the minimum criteria described above. Applying *Fisher*'s [1953] statistics yields a mean VGP of 87.4°N, 162.9°E, and α_{95} = 3.9° (Table 2). This mean VGP is nearly coincident with the rotational axis (Figure 7a), and the total angular dispersion, S_T, about the mean pole is 18.3°. The dispersion, S_F, corrected for within-site dispersion due to experimental errors as estimated using the method of *McElhinny and Merrill* [1975] is 17.9°. The upper and lower 95% confidence limits for S_F are equal to 20.3° and 16.0°, respectively, as calculated using the method of *Cox* [1969]. Bingham statistics [*Onstott*, 1980] show that the data are circularly distributed (Figure 7a), i.e., the axial ratio is not significantly different at 95% confidence [*Schmidt*, 1990]. The mean normal- and reversed-polarity poles differ by 169.4° (Figure 7b) and are not antipodal at greater than 95% confidence [*McFadden and Lowes*, 1981]. Asymmetry of normal and reversed polarities over the last 5 million years has been long known [e.g., *Cox*, 1975; *Merrill and McElhinny*, 1977, 1983; *Schneider and Kent*, 1988; *Merrill et al.*, 1990] but the cause is uncertain. Incomplete removal of secondary components, especially from reversed-polarity rocks, is often suspected. This explanation does not seem entirely satisfactory here; note that the 95% confidence

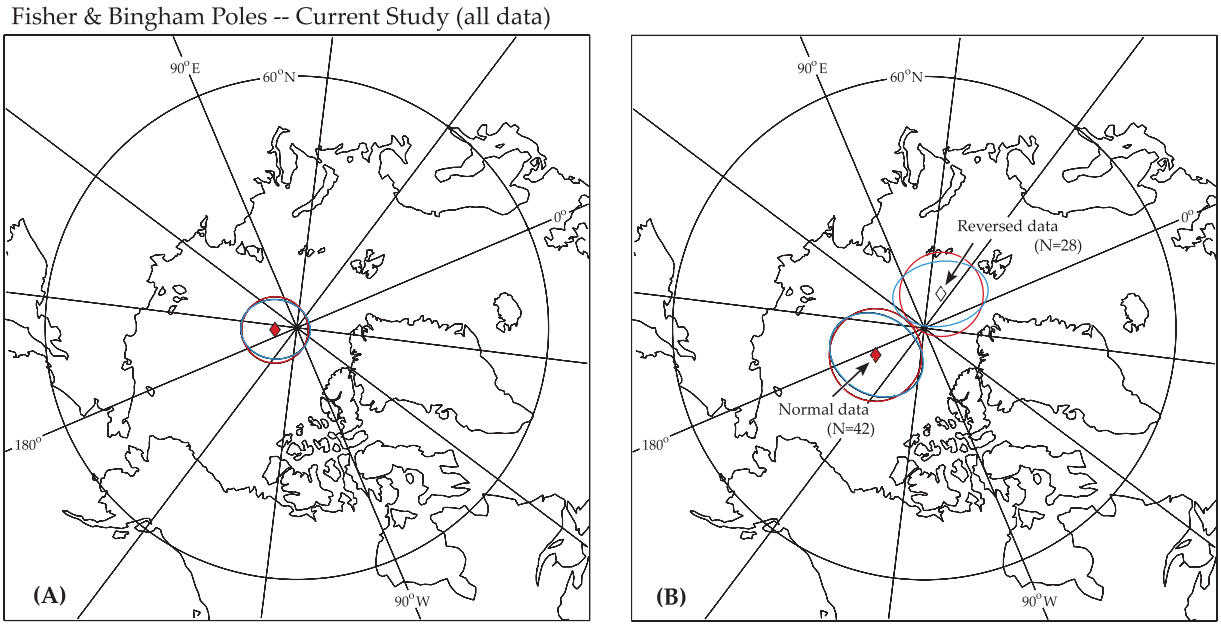


Figure 7. (a) Mean VGP (diamond) for 70 sites from the current study and (b) mean VGPs for normal- and reversed-polarity sites. *Fisher* [1953] and Bingham [*Onstott*, 1980] 95% confidence ovals are shown in red and blue, respectively.

circle about the mean reversed-polarity data includes the rotational axis whereas that for the normal-polarity data does not (Figure 7b). The dispersion of the normal data set also is considerably higher than that of the reversed (Table 2). The polarity asymmetry may be due to unspecified conditions at the core-mantle boundary that have varied slowly in time and space.

4.2. Previous Studies

[34] To carry this analysis further, I include those studies in the southwestern United States that were published by researchers at the USGS Menlo Park laboratory. Any differences, however small, that might be due to different techniques or instrumentation between laboratories should be eliminated with this restriction. The minimum criteria described above are applied and no NRM data included in the published works are used here. I also restrict paleomagnetic data to the same age range as for the current study (<10 Ma). Included in this expanded data set are 4 Brunhes-age and 6 Matuyama-age units from Valles caldera, New Mexico [*Doell et al.*, 1968], and 20 Brunhes-age units from Long Valley caldera, California [*Mankinen et al.*, 1986]. The mean pole reported by *Palmer et al.* [1996] for the Bishop Tuff is used. Studies in the Clear Lake volcanic field, California [*Mankinen et al.*, 1978, 1981] were designed to define the limits of the Jaramillo and Cobb Mountain Subchrons and thus many units sampled had transitional directions. Four Matuyama-age units from those studies can be included here, as can be a fifth unit from the Clear Lake field (site 31) sampled by *Mankinen* [1972]. Results from two Matuyama-age units in the Coso volcanic field were reported by *Mankinen and Grommé* [1982]. Results from thirty-six sites in the Miocene Stanislaus Group of the central Sierra Nevada were reported by *King et al.* [2007]. These units include four lava flows in the Dardanelles Formation, six in the Table Mountain Latite, and single ash flow tuffs of the Tollhouse Flat, By-Day, and Upper Members of the Eureka Valley Tuff were sampled. Eleven of these units are reliable and thus included in this analysis.

[35] The final group of rocks to be considered is from the Sonoma Volcanics, California [*Mankinen*, 1972, 1989]. The paleomagnetic pole calculated for 27 cooling units proved to be anomalous (significantly displaced from the geographic pole) despite positive fold and reversal tests. Twenty-two of the units erupted during the Gilbert Chron, one during the Gauss, and four during late Miocene time. A somewhat elliptical VGP distribution [*Mankinen*, 1989] led to the conclusion that geomagnetic secular variation had not been averaged despite the range in ages and geographical distri-

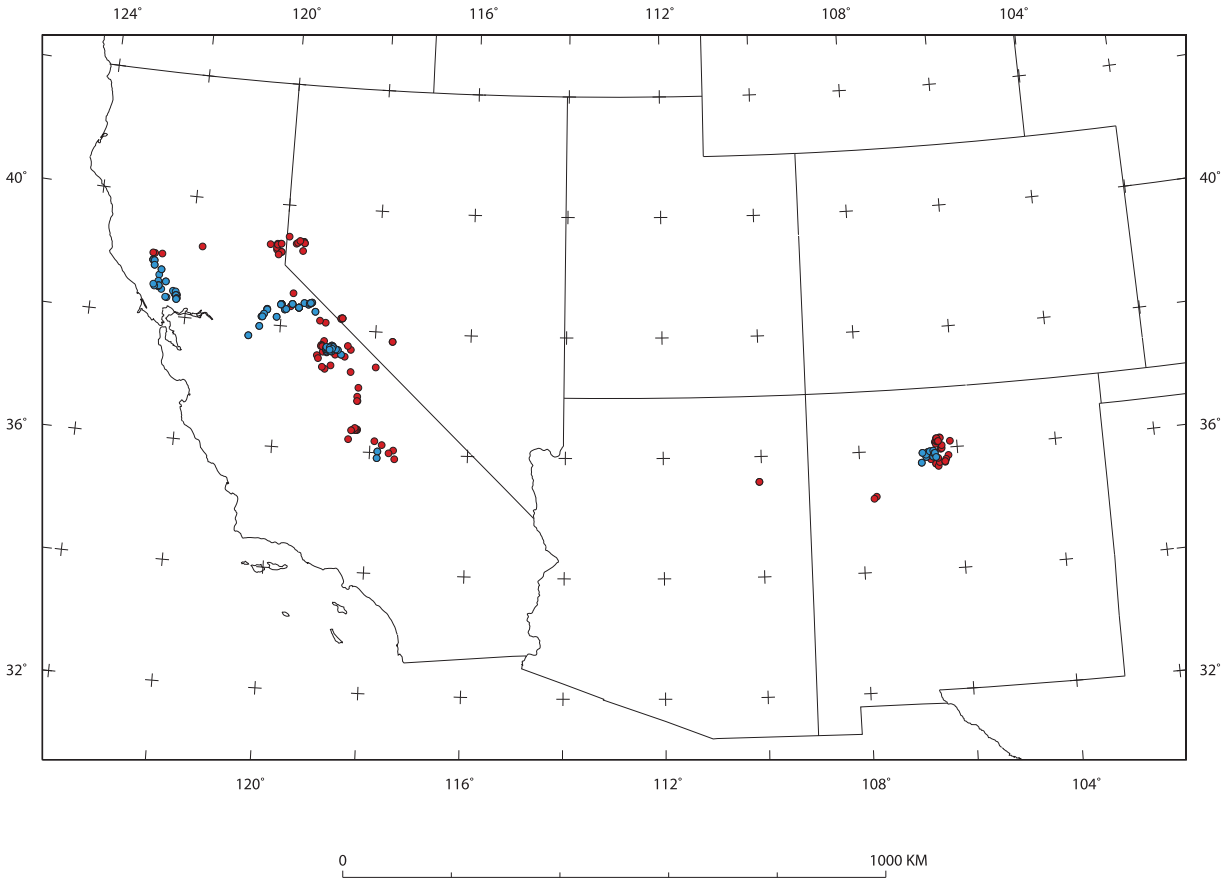


Figure 8. Index map showing the region studied with red dots denoting sites described in the current study and cyan dots denoting sites previously published (see text).

bution of sample sites. Although this result is highly unusual for young volcanic rocks, it is not unique [*Sheriff*, 1989]. *Tauxe et al.* [2003] suggested that inappropriate tilt corrections may have been responsible for the discrepant result. However, structural attitudes in many cases were determined on sedimentary interbeds, the data yield a positive tilt test [*Enkin*, 2003], and the majority of sites were sampled northeast of the major faults of the area. If a few of the most discrepant VGPs were the result of poor structural control, the total distribution could be more circular although an inspection of Figure 2 of *Mankinen* [1989] shows that almost all poles are displaced to the far-side of the geographic axis. A uniform tilting of a crustal block of this size (more than 2,000 km^2) seems unlikely as does the possibility of more than 300 km of northward translation for rocks this young. Because there does not seem to be an obvious justification for disregarding the results from the Sonoma Volcanics, they will be included here with the assumption that a less anomalous distribution will be apparent when they are combined with results from other rocks of similar age.

4.3. Combined Results

[36] Paleomagnetic results are available from 146 localities (Figure 8) in the revised data set. These data yield a mean VGP of 86.3°N, 107.6°E, α_{95} = 2.8°, and S_T = 19.1° (Table 2). The distribution of VGPs again is circular but significantly displaced from the rotational axis (Figure 9a). Similar to the current data set, the mean normal- and reversed-polarity poles differ by 172.8° and are not antipodal at greater than 95% confidence (Figure 9b). Here, however, the reversed data have an elliptical VGP distribution that is significantly different at 95% confidence [*Schmidt*, 1990]. Separating the data into specific age groups, I find that the *Fisher* [1953] 95% confidence limits about the Brunhes and Matuyama paleomagnetic poles include the rotational axis (Figures 10a and 10b). Only 17 data were available from the Gauss Chron, so data from the Gauss and Gilbert Chrons were combined and

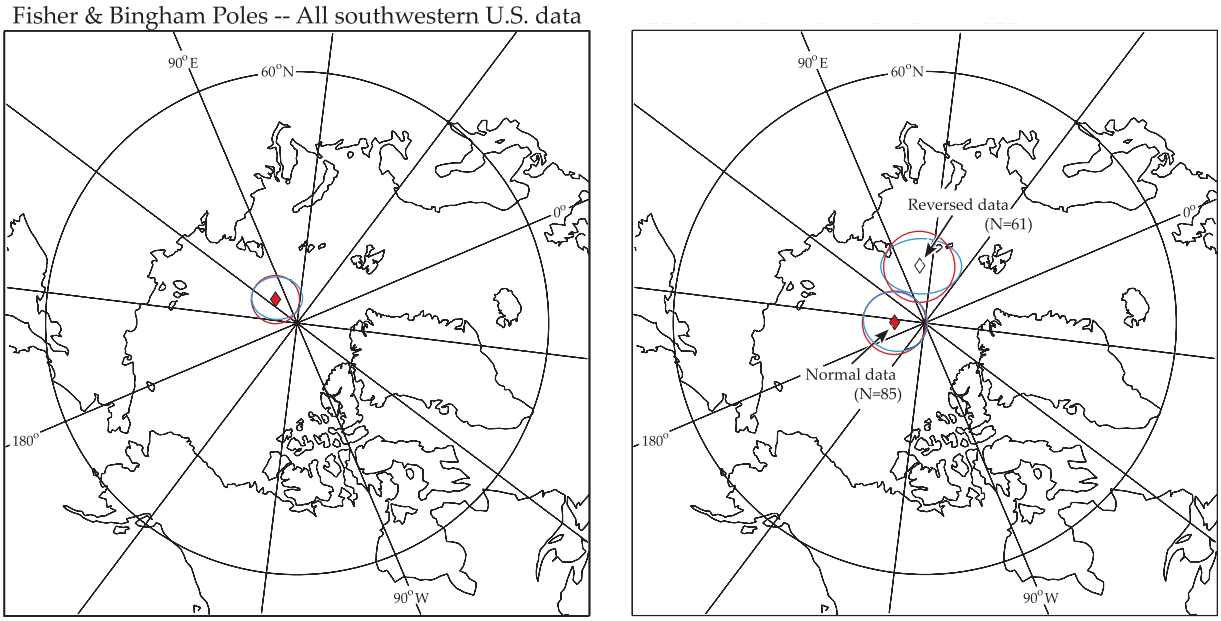


Figure 9. (a) Mean VGP (diamond) for 146 sites and (b) mean VGPs for normal- and reversed-polarity sites. Results combine data from the current study with those previously published (see text). See Figure 7 for additional explanation.

the 95% confidence oval for this mean pole again includes the rotational axis (Figure 10c) as does the late Miocene (~10 to 5 Ma) pole (Figure 10d). The 95% confidence oval about the late Miocene pole is relatively large (7.8°), probably because there are only 30 results from this 5 Ma period of time as compared with 116 results for the Gilbert through Brunhes interval. This late Miocene pole seems reasonably accurate, however, because it does not differ significantly from either the Steens Basalt (~16 Ma) or the time-equivalent Columbia River Basalt Group poles [*Mankinen et al.*, 1987]. Because this time period is older than that being considered by the Time-Averaged Field Initiative (TAFI, see *Johnson et al.* [2005]) it will not be considered further here. When all Miocene data are excluded, the mean paleomagnetic pole is within 3° of the rotational axis (PLA = 87.1°, PLO = 97.8°, α_{95} = 3.0°).

4.4. Paleosecular Variation

[37] Statistical parameters for all the volcanic units analyzed are given in Table 2. S_T is the total angular dispersion about the mean value for each of the groups as described above. Another common practice is to calculate the total dispersion with respect to the geographic axis, as was done, for example, by *McFadden et al.* [1988, 1991]. Because the geologically young igneous rocks analyzed here give indications of having successfully averaged secular variation, however, there should be little difference. Indeed, the total dispersion about the mean pole for the 146 sites in the combined data set is 19.1°, whereas the dispersion about the geographic axis is 19.5°. Dispersions calculated both ways differ by substantially less than 1° for most of the groups in Table 2 except for the Miocene data where the difference is 1.2°. Direct comparison with other studies is therefore reasonable using S_F (the dispersion corrected for within-site dispersion due to experimental errors), with upper and lower 95% confidence limits as calculated using the method of *Cox* [1969]. It is apparent (Table 2) that the dispersion about the mean paleomagnetic poles increases systematically with time. The dispersions for Brunhes and Matuyama time are consistent with S_F = 15.5° (upper and lower limits of 16.6° and 14.5°, respectively) in the 30°–40° latitude range of model G [*McFadden et al.*, 1988, 1991] for the past 5 Ma. The dispersions for Gauss/Gilbert and late Miocene time are consistent with model G for the 5 to 22.5 Ma time range, assuming that S_F for the 30°–40° latitude range is ~20° [see *McFadden et al.*, 1991, Table 3]. Thus, the increase in dispersion with time shown here is entirely consistent with the

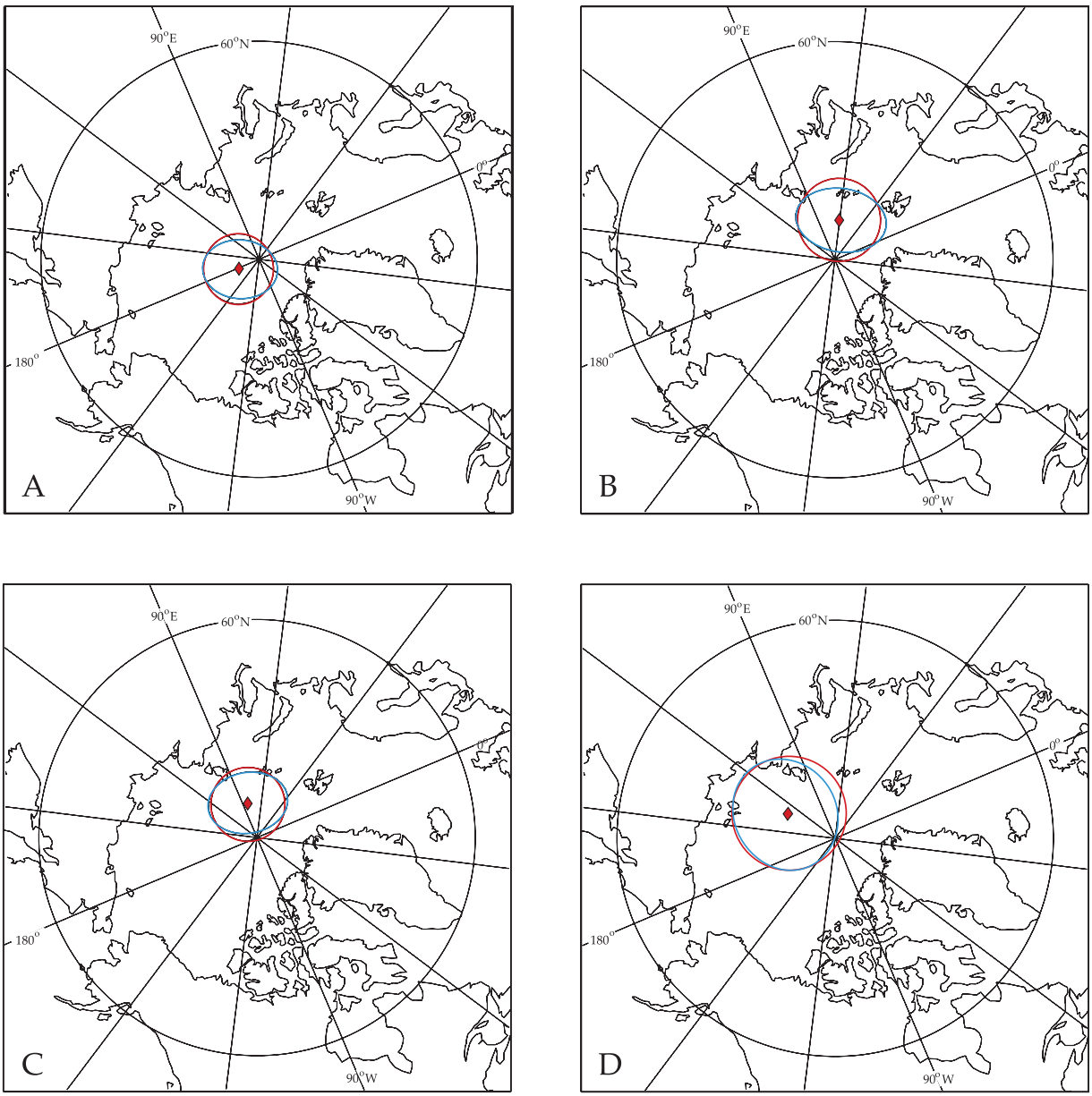


Figure 10. Mean VGPs (diamonds) and associated 95% confidence ovals for data from (a) the Brunhes, (b) Matuyama, (c) Gauss/Gilbert, and (d) late Miocene. See Figure 7 for additional explanation.

predictions of paleosecular variation model G of *McFadden et al.* [1988, 1991].

5. Conclusions

[38] Paleomagnetic samples, for which complete data were not previously published, were obtained from igneous rocks at 96 localities in the southwestern United States. Ages range from late Miocene (~10 Ma) to late Pleistocene (~10 ka). Of this group, 70 sites provided results that meet certain minimum criteria and can be considered for the Time-Averaged Field Initiative (TAFI). The virtual geomagnetic poles for these 70 units are circularly distributed and their mean is nearly coincident with the rotational axis (Figure 7). The mean normal- and reversed-polarity poles, however, differ by 169.4° and are not antipodal at greater than 95% confidence. When all the published data for southwestern USA that were measured at the USGS laboratory are included

(N = 146), the VGPs are again circularly distributed but the mean is significantly displaced from the rotational axis (Figure 9). The mean normal- and reversed-polarity poles of this group differ by 172.8° and again are not antipodal at greater than 95% confidence. When the data are separated into specific age groups, the 95% confidence limits about the Brunhes, Matuyama, Gauss/Gilbert, and late Miocene paleomagnetic poles all include the rotational axis (Figure 10). The increasing dispersions about the mean paleomagnetic poles with age are consistent with paleosecular variation model G of *McFadden et al.* [1988, 1991].

Appendix A: Site Locations and Descriptions

[39] Following are locations of sites sampled for paleomagnetic study by personnel of the U.S. Geological Survey Rock Magnetics Laboratory (Menlo Park, CA) for which complete results were not published. In a few instances, original field maps could not be located and site descriptions were reconstructed from field notes and recorded site coordinates. Included in the site descriptions are ages determined for the individual units. Many ages originally published were calculated using atomic abundance and decay constants for ^{40}K that were subsequently revised [*Steiger and Jäger*, 1977]. K-Ar ages reported here were recalculated using the revised constants [*Mankinen and Dalrymple*, 1979; *Dalrymple*, 1979] whenever necessary. A five digit flow identifier precedes each description, with a letter denoting the researcher who collected the samples (C & V = Cox, D & X = Doell, G = Dalrymple). Reference numbers in the descriptions are keyed to Table A1.

A1. California-Nevada

[40] 1C118: lat 37.06°N.; long 241.70°E. Big Pine, Calif. 1:62500 quadrangle (1950). Rhyolite flow in NE¼ sec. 25, T. 10 S., R. 33 E., and N½ sec. 30, T. 10 S., R. 34 E. Reversed polarity. Age = 1.01 ± 0.04 Ma. Polarity and age ref. 1, unit S2.

[41] 1C138: lat 36.84°N.; long 241.70°E. Mt. Pinchot, Calif. 1:62500 quadrangle (1953). Basalt flow above canyon floor of North Fork Oak Creek near center of top boundary SW¼ sec. 5, T. 13 S., R. 34 E. Reversed polarity.

[42] 1C143: lat 37.41°N.; long 242.04°E. Soldier Pass, Calif.-Nev. 1:62500 quadrangle (1958). Basalt flow on south side of Route 63 at north end of Deep Springs Valley. S½NW¼ sec. 19, T. 6 S., R. 37 E. Reversed polarity (ref. 2, unit W35). Age = 11.1 ± 1.0 Ma (ref. 3, sample KA968).

Table A1. Sources of Information for Site Descriptions in Appendix A

Data Source Number	Reference
1	*Cox et al.* [1963a]
2	*Dalrymple and Hirooka* [1965]
3	*Dalrymple* [1963]
4	*Cox et al.* [1963b]
5	*Dalrymple* [1964b]
6	*Dalrymple et al.* [1982]
7	*Dalrymple et al.* [1967]
8	*Dalrymple* [1980]
9	*Huber and Rinehart* [1967]
10	*Donnelly-Nolan et al.* [1981]
11	*Evernden et al.* [1957]
12	*Cox et al.* [1964]
13	*Silberman et al.* [1979]
14	*Doell et al.* [1966]
15	*Dalrymple* [1964a]
16	*Moore and Lanphere* [1983]
17	*Moore and Sisson* [1985]
18	*Dalrymple and Lanphere* [1971]
19	*Duffield et al.* [1980]

[43] 1C151: lat 37.85°N.; long 242.36°E. Silverpeak, Nev. 1:62500 quadrangle (1963). Basalt flow on west side of "The Monocline" north of Silverpeak. Sampled in SW¼ sec. 15, T. 1 S., R. 39 E. Normal polarity.

[44] 1C160: lat 37.85°N.; long 242.36°E. Silverpeak, Nev. 1:62500 quadrangle (1963). Basalt flow underlying unit 1C151 at same locality. Normal polarity.

[45] 1C164: lat 38.33°N.; long 240.19°E. Dardanelles Cone, Calif. 1:62500 quadrangle (1956). Basalt flow south of Middle Fork Stanislaus River near center of SE¼ sec. 30, T. 6 N., R. 20 E. Normal polarity (ref. 4, unit S8). Age = 0.158 ± 0.025 Ma (refs. 4 and 5).

[46] 1C601: lat 37.58°N.; long 241.18°E. Mt. Morrison, Calif. 1:62500 quadrangle (1953). McGee Mtn. basalt overlain by McGee Till near center of western part of SW¼ sec. 19, T. 4 S., R. 29 E. Normal polarity (ref. 1, unit S5). Age = 2.71 ± 0.10 Ma (ref. 3, sample KA973).

[47] 1C651: lat 37.80°N.; long 240.93°E. Mono Craters, Calif. 1:62500 quadrangle (1953). Basalt flow near June Lake in SW¼NW¼ sec. 1, T. 2 S., R. 26 E. Normal polarity.

[48] 1C671: lat 37.68°N.; long 241.50°E. Casa Diablo Mtn, Calif. 1:62500 quadrangle (1953).

Basalt flow on east side of ridge in SE¼ sec. 13, T. 3 S., R. 31 E. Normal polarity.

[49] 1C684: lat 36.91°N.; long 241.69°E. Mt. Pinchot, Calif. 1:62500 quadrangle (1953). Basalt flow between forks of Sawmill Creek on "The Hogback." 1.6 km west of western boundary of sec. 17, T. 12 S., R. 34 E. Normal polarity (ref. 4, unit S7). Age = 0.053 ± 0.044 Ma (refs. 4, 5, and 6).

[50] 1C701: lat 36.84°N.; long 241.69°E. Mt. Pinchot, Calif. 1:62500 quadrangle (1953). Basalt flow along North Fork Oak Creek sampled where Baxter Pass Trail crosses the stream near western boundary sec. 5, T. 13 S., R. 34 E. Normal polarity.

[51] 2C508: lat 37.31°N.; long 241.52°E. Bishop, Calif. 1:62500 quadrangle (1949). Basalt of Coyote Flat sampled along unimproved road at elevation 6539′ in SE¼ sec. 29, T. 7 S., R. 32 E. Reversed polarity (ref. 7, unit W34). Age = 9.9 ± 0.2 Ma (ref. 3, sample KA996).

[52] 2C516: lat 37.56°N.; long 241.38°E. Casa Diablo Mtn, Calif. 1:62500 quadrangle (1953). Four flow units of basalt of Owens Gorge sampled in NW¼ sec. 36, T. 4 S., R. 30 E. Intermediate polarity. Age = 3.35 ± 0.10 Ma (ref. 3, sample KA1013). Originally reported to possess a normal magnetic polarity — see ref. 1, unit S6.

[53] 2C551: lat 37.62°N.; long 240.92°E. Devils Postpile, Calif. 1:62500 quadrangle (1953). Basalt at Devils Postpile sampled 945 meters S35°E from BM 7559. 0.4 km east of eastern boundary of Devils Postpile National Monument. Stratigraphically overlies the Bishop Tuff. Normal polarity (ref. 4, unit S9). Different plagioclase separates from the same sample have yielded ages of 0.97 ± 0.16 Ma (refs. 4 and 5) and 0.65 + 0.35 Ma (ref. 9).

[54] 2C576: lat 37.72°N.; long 240.89°E. Devils Postpile, Calif. 1:62500 quadrangle (1953). Basalt of San Joaquin Mtn. sampled 0.7 km NW of VABM 11600 at 10,640′ elevation. Normal polarity. Age = 3.23 ± 0.10 Ma. Polarity and age ref. 4, unit S16.

[55] 2C582: lat 37.73°N.; long 240.88°E. Devils Postpile, Calif. 1:62500 quadrangle (1953). Basalt of San Joaquin Mtn. sampled near center of eastern edge of SE¼ sec. 32, T. 2 S., R. 26 E. Normal Polarity.

[56] 2C591: lat 37.71°N.; long 240.87°E. Devils Postpile, Calif. 1:62500 quadrangle (1953). Near bottom of section of basalt flows on San Joaquin Mtn. and 1.6 km due east of Altha Lake. Normal polarity.

[57] 2C598: lat 37.71°N.; long 240.90°E. Devils Postpile, Calif. 1:62500 quadrangle (1953). Oriented hand-samples of quartz latite collected from the headwall of northwest-facing cirque on Two Teats Mountain. Stratigraphically overlies unit 2C576. Normal polarity. Age = 3.09 ± 0.10 Ma. Polarity and age ref. 4, unit S15.

[58] 2C801: lat 39.01°N.; long 237.24°E. Lakeport, Calif. 1:62500 quadrangle (1958). Dacite of Horseshoe Bay sampled on south side of Horseshoe Bend in sec. 5, T. 13 N., R. 8 W. Normal polarity. Age = 0.35 ± 0.02 Ma (ref. 10, unit dh).

[59] 2C804: lat 39.01°N.; long 237.20°E. Lakeport, Calif. 1:62500 quadrangle (1958). Dacite flow on west side of Soda Bay near western boundary of sec. 6, T. 13 N., R. 8 W. Normal polarity. Age = 0.35 ± 0.06 Ma (ref. 10, unit dsb).

[60] 2C809: lat 39.01°N.; long. 237.40°E. Clearlake Oaks, Calif. 1:62500 quadrangle (1960). Basalt flow approximately 150 m east of Route 20 near southern border of sec. 35, T. 14 N., R. 7 W. Reversed polarity.

[61] 2C821: lat 39.19°N.; long 238.23°E. Sutter Buttes, Calif. 1:24000 quadrangle (1954). Rhyolite sampled on SE side of hill 607 in SW¼ sec. 32, T. 16 N., R. 2 E. approximately 0.5 km north of BM 168 on Pass Road. Reversed polarity (ref. 1, unit S4). Age = 1.74 ± 0.10 Ma (ref. 11, sample KA 65).

[62] 2D001: lat 39.35°N.; long 239.81°E. Truckee, Calif.-Nev. 1:62500 quadrangle (1955). Basalt near top of Alder Hill along north edge SW¼ sec. 3, T. 18 N., R. 16 E. northwest of Truckee. Normal polarity. Ref. 1, unit S14. Age originally reported for the unit is in error; see ref. 12.

[63] 2D005: lat 39.35°N.; long 239.88°E. Truckee, Calif.-Nev. 1:62500 quadrangle (1955). Basalt flow along south edge NW¼ sec. 5, T. 17 N., R. 17 E. on east bank of Truckee River and 0.3 km north of Martis Creek. Reversed polarity.

[64] 2D010: lat 39.48°N.; long 240.05°E. Mt. Rose, Nev. 1:62500 quadrangle (1952). Basalt flow of Lousetown Formation sampled near center of eastern edge sec. 28, T. 19 N., R. 18 E. Normal polarity.

[65] 2D018: lat 39.41°N.; long 240.37°E. Virginia City, Nev. 1:62500 quadrangle (1952). Lowest flow of the lower basalts of the Lousetown For-

mation near center of south edge sec. 21, T. 18 N., R. 21 E. 0.5 km north of BM 5701 on Lousetown Creek. Reversed polarity.

[66] 2D025: lat 39.41°N.; long 240.37°E. Virginia City, Nev. 1:62500 quadrangle (1952). Basalt overlying unit 2D018 at same locality. Intermediate polarity. Age = 7.08 ± 0.19 Ma (ref. 7, unit W22).

[67] 2D031: lat 39.39°N.; long 240.39°E. Virginia City, Nev. 1:62500 quadrangle (1952). Basalt of Lousetown Formation sampled on northwest side of ephemeral lake near center of eastern edge SW¼ sec. 27, T. 18 N., R. 21 E. Intermediate polarity.

[68] 2D036: lat 39.39°N.; long 240.39°E. Virginia City, Nev. 1:62500 quadrangle (1952). Basalt flow overlying unit 2D031 at same locality. Intermediate polarity.

[69] 2D041: lat 39.37°N.; long 240.21°E. Mt. Rose, Nev. 1:62500 quadrangle (1952). Steamboat Hills rhyolite sampled near center of top boundary SE¼ sec. 1, T. 17 N., R. 19 E. Reversed polarity. Age = 1.14 ± 0.05 Ma (ref. 13).

[70] 2D045: lat 39.39°N.; long 240.24°E. Mt. Rose, Nev. 1:62500 quadrangle (1952). Basaltic andesite of the Steamboat Hills in NW¼ sec. 32, T. 18 N., R. 20 E. at 5,060′ elevation and 0.1 km SE of clay pit. Reversed polarity.

[71] 2D051: lat 39.26°N.; long 240.36°E. Virginia City, Nev. 1:62500 quadrangle (1952). Basalt of McClellan Peak capping small hill in SE¼ sec. 8, T. 16 N., R. 21 E. North side of American Ravine and west of Silver City. Reversed polarity. Age = 1.16 ± 0.04 Ma. Polarity and age ref. 14, unit S18.

[72] 2D056: lat 39.41°N.; long 240.29°E. Virginia City, Nev. 1:62500 quadrangle (1952). Perlite rib on Steamboat Hills Rhyolite dome in NW¼ sec. 23, T. 18 N., R. 20 E. on east side of Truckee Meadows. 3.1 km N70°E from Browns School. Intermediate polarity.

[73] 2D066: lat 39.25°N.; long 239.81°E. Truckee, Calif.-Nev. 1:62500 quadrangle (1955). Big Chief basalt near center of south edge NW¼ sec. 10, T. 16 N., R. 16 E. at 6,900′ elevation. 0.9 km S35°W from 7,557′ elevation on Sawtooth Ridge. Reversed polarity.

[74] 2D072: lat 39.28°N.; long 239.80°E. Truckee, Calif.-Nev. 1:62500 quadrangle (1955). Bald Mtn. olivine latite in NE¼ sec. 33, T. 16 N., R. 16 E. at 6,250′ elevation. 0.7 km S68°E from BM 5898 on Hwy 89. Reversed polarity. Age = 1.28 ± 0.10 Ma. Polarity and age ref. 4, unit S10.

[75] 3D101: lat 39.22°N.; long 239.91°E. Tahoe, Calif.-Nev. 1:62500 quadrangle (1955). Andesite flow west side of Hwy 28 in SE¼ sec. 21, T. 16 N., R. 17 E. approximately 90 meters south of Watson Creek. Reversed polarity. Age = 2.23 + 0.10 Ma (ref. 4, unit S13). Originally reported to possess a normal magnetic polarity, but samples used for the polarity determination were later found to have come from underlying unit 3D107; see ref. 12.

[76] 3D107: lat 39.22°N.; long 239.91°E. Tahoe, Calif.-Nev. 1:62500 quadrangle (1955). Andesite flow on north side of Watson Creek along Hwy 28 in SE¼ sec. 21, T. 16 N., R. 17 E. Normal polarity (ref. 14, unit S20). Age = 2.52 ± 0.07 Ma (refs. 14 and 15).

[77] 3D115: lat 39.17°N.; long 239.85°E. Tahoe, Calif.-Nev. 1:62500 quadrangle (1955). Olivine latite flow near Tahoe City. North of Truckee River in NE¼ sec. 12, T. 15 N., R. 16 E. at 6,680′ elevation. Reversed polarity. Age = 1.93 ± 0.10 Ma. Polarity and age ref. 4, unit S12.

[78] 3D131: lat 39.33°N.; long 239.82°E. Truckee, Calif.-Nev. 1:62500 quadrangle (1955). Olivine latite flow near Polaris in SE¼ sec. 11, T. 17 N., R. 16 E. and 0.4 km east of gravel pit on north side of Hwy 40 east of Truckee. Reversed polarity. Age = 1.68 ± 0.05 Ma. Polarity and age ref. 14, unit S19.

[79] 3D138: lat 39.35°N.; long 239.89°E. Truckee, Calif.-Nev. 1:62500 quadrangle (1955). Olivine latite flow near Hirshdale. East side NW¼ sec. 5, T. 17 N., R. 17 E. South of Hwy 40 and approximately 0.2 km east of intersection with road to Union Mills. Reversed polarity. Age = 1.29 ± 0.10 Ma. Polarity and age ref. 4, unit S11.

[80] 3V001: lat 37.40°N.; long 241.10°E. Mt. Abbot, Calif. 1:62500 quadrangle (1953). Basalt flow 0.5 km west of Volcanic Knob. Reversed polarity (ref. 14, unit S27). Age = 3.66 ± 0.10 Ma (ref. 3, sample KA988).

[81] 3V012: lat 37.34°N.; long 240.98°E. Kaiser Peak, Calif. 1:62500 quadrangle (1953). Basalt flow sampled on NE side of Devils Table near BM 7222 and 1.5 km NW of Mono Hot Springs. Reversed polarity (ref. 14, unit S24). Age = 3.55 ± 0.10 Ma (ref. 3, sample KA997).

[82] 3V021: lat 36.36°N.; long 241.72°E. Kern Peak, Calif. 1:62500 quadrangle (1956). Lava flow from South Fork cone. North side of South Fork Kern River in SW¼ of NE¼ sec. 15, T. 18 S., R. 34 E. Normal polarity. Unpublished K-Ar age = 0.176 ± 0.021 Ma (refs. 16 and 17).

[83] 3V028: lat 36.37°N.; long 241.72°E. Kern Peak, Calif. 1:62500 quadrangle (1956). Lava flow from Tunnel cone near center of north boundary of sec. 15, T. 18 S., R. 34 E. Sampled at top of ridge approximately 0.65 km east of Tunnel Guard Station. Normal polarity.

[84] 3V038: lat 36.36°N.; long 241.70°E. Kern Peak, Calif. 1:62500 quadrangle (1956). Lava flow from South Fork cone on north side of Golden Trout Creek in NW¼SE¼ sec. 16, T. 18 S., R. 34 E. Normal polarity.

[85] 3V048: lat 36.36°N.; long 241.68°E. Kern Peak, Calif. 1:62500 quadrangle (1956). Postglacial basalt flow forming east and west rims of Groundhog cone NE of Volcano Meadow near center of sec. 17, T. 18 S., R. 34 E. Normal polarity.

[86] 3V061: lat 36.36°N.; long 241.63°E. Kern Peak, Calif. 1:62500 quadrangle (1956). South Fork cone basalt flow approximately 150 meters north of Natural Bridge and 2.6 km west of west boundary of sec. 18, T. 18 S., R. 34 E. Normal polarity.

[87] 3V069: lat 36.39°N.; long 241.67°E. Kern Peak, Calif. 1:62500 quadrangle (1956). Lava flow from Little Whitney cone. Sampled on ridge between Johnson Creek and creek from Salt Lick Meadow in SE¼ sec. 6, T. 18 S., R. 34 E. Reversed polarity. Unpublished K-Ar age = 0.743 ± 0.011 Ma (refs. 16 and 17).

[88] 3V077: lat 36.35°N.; long 241.60°E. Kern Peak, Calif. 1:62500 quadrangle (1956). South Fork cone lava flow on east side of Kern Canyon along trail to Volcano Falls. 1.25 km NE of Kern Canyon Ranger Station. Normal polarity.

[89] 3V085: lat 36.20°N.; long 241.55°E. Hockett Peak, Calif. 1:62500 quadrangle (1956). Basalt of Trout Meadows sampled SE of confluence of Little Kern River and Deep Creek. Reversed polarity (ref. 14, unit S26). Age = 3.56 ± 0.10 Ma (ref. 3, sample KA1014).

[90] 3V093: lat 37.37°N.; long 240.92°E. Kaiser Peak, Calif. 1:62500 quadrangle (1953). Basalt flow north of South Fork San Joaquin River and 5.8 km west of Lake Thomas A. Edison. North side of small hill at 7.040′ elevation and south of trail between Warm Creek and Four Forks Creek. Reversed polarity (ref. 14, unit S28). Age = 3.57 ± 0.10 Ma (refs. 14 and 15).

[91] 3V101: lat 37.56°N.; long 240.80°E. Devils Postpile, Calif. 1:62500 quadrangle (1953). Basalt flow on north side of trail between Sheep Crossing and Snake Meadow near BM 6711. Normal polarity (ref. 14, unit S23). Age = 3.35 ± 0.10 Ma (ref. 15, sample KA1187).

[92] 3V109: lat 37.51°N.; long 240.83°E. Devils Postpile, Calif. 1:62500 quadrangle (1953). Basalt near Pine Flat and 0.8 km S65;E from NE corner sec. 12, T. 5 S., R. 25 E. at 6,700′ elevation. Reversed polarity (ref. 14, unit S25). Age = 3.60 ± 0.10 Ma (ref. 15, sample KA1186).

[93] 3V121: lat 37.74°N.; long 241.43°E. Casa Diablo Mtn, Calif. 1:62500 quadrangle (1953). Benton Range basalt sampled near eastern edge NE¼ sec. 32, T. 2 S., R. 31 E. at 7,900′ elevation. Normal polarity (ref. 14, unit S22). Age = 3.40 ± 0.07 Ma (refs. 9, 14, 15, and 18).

[94] 3V129: lat 38.18°N.; long 241.26°E. Huntoon Valley, Nev.-Calif. 1:62500 quadrangle (1958). Biotite-augite latite tuff in T. 4 N., R. 29 E. southwest of Anchorite Pass. NE side of Hwy 31 and approximately 0.15 km SW of BM 7257. Reversed polarity. Age = 9.77 ± 0.38 Ma. Polarity and age ref. 7, unit W23.

[95] 3V138: lat 38.10°N.; long 240.94°E. Bodie, Calif. 1:62500 quadrangle (1958). Biotite-augite latite tuff near center of southern edge SW¼ sec. 24, T. 3 N., R. 26 E. North of road and approximately 1.6 km SW of Goat Ranch. Reversed polarity. Age = 9.43 ± 0.28 Ma. Polarity and age ref. 7, unit W24.

[96] 3V148: lat 37.61°N.; long 241.00°E. Mt. Morrison, Calif. 1:62500 quadrangle (1953). Basalt flow in SW¼SE¼ sec. 9, T. 4 S., R. 27 E. approximately 300 m south of Old Mammoth mine. Reversed polarity (ref. 14, unit S21). Age = 3.11 ± 0.02 Ma (refs. 14 and 15).

[97] 3V156: lat 38.13°N.; long 240.82°E. Bodie, Calif. 1:62500 quadrangle (1958). Biotite-augite latite tuff sampled on both sides of Hwy 395 in SE¼ sec. 11, T. 3 N., R. 25 E. Approximately 0.25 km south of BM 7594. Normal polarity. Age = 9.42 ± 0.22 Ma. Polarity and age ref. 7, unit W25.

[98] 3V167: lat 38.55°N.; long 240.22°E. Markleeville, Calif. 1:62500 quadrangle (1956). Rhyolite of Nobel Canyon in NE¼ sec. 16 (unsurveyed), T. 8 N., R. 20 E. 2.1 km S49 W from VABM 10774 on Silver Peak at 8,100′ elevation. Reversed polarity (ref. 7, unit W30). Age = 4.87 ± 0.07 Ma (ref. 15, samples KA1119 and KA1120).

[99] 3V176: lat 39.33°N.; long 239.66°E. Donner Pass, Calif. 1:62500 quadrangle (1955). Basalt of Boreal Ridge. Approximately 0.3 km due east of northwest corner of sec. 17, T. 17 N., R. 15 E. at 7,600′ elevation. Normal polarity. Age = 7.6 ± 0.2 Ma. Polarity and age ref. 7, unit W31.

[100] 3X901: lat 36.19°N.; long 242.09°E. Haiwee Reservoir, Calif. 1:62500 quadrangle (1951). Rhyolite sampled near center of south edge SW¼ sec. 13, T. 20 S., R. 37 ½E. at 5,160′ elevation. Reversed polarity.

[101] 3X906: lat 35.91°N.; long 242.51°E. Trona, Calif. 1:62500 quadrangle (1949). Basalt sampled in Argus Range approximately 7 km NW of Argus Peak. Near center sec. 30, T. 23 S., R. 42 E. at 6,200′ elevation and immediately SW of elevation 6268. Reversed polarity.

[102] 3X914: lat 36.05°N.; long 242.48°E. Coso Peak, Calif. 1:62500 quadrangle (1951). Flow near center sec. 2, T. 22 S., R. 41 E. west of Etcheron Valley. North side of road at 5,920′ elevation. Reversed polarity.

[103] 3X921: lat 36.00°N.; long 242.39°E. Mountain Springs Canyon, Calif. 1:62500 quadrangle (1953). Flow just north of center sec. 25, T. 22 S., R. 40 E. at end of road at 5,000′ elevation. Reversed polarity.

[104] 5G102: lat 38.19°N.; long 241.29°E. Huntoon Valley, Nev.-Calif. 1:62500 quadrangle (1958). Perlite flow in T. 4 N., R. 29 E. southwest of Anchorite Pass. North side of Hwy 31 and approximately 0.4 km N32°E from BM 7401. Reversed polarity. Age = 3.43 ± 0.11 Ma. Polarity and age ref. 7, unit W19.

[105] 5G113: lat 38.19°N.; long 241.29°E. Huntoon Valley, Nev.-Calif. 1:62500 quadrangle (1958). Biotite andesite flow in T. 4 N., R. 29 E. southwest of Anchorite Pass. North side of Hwy 31 at 7,720′ elevation and 0.6 km N2°W from BM 7401. Reversed polarity. Age = 3.71 ± 0.18 Ma. Polarity and age ref. 7, unit W20.

[106] 5G122: lat 38.19°N.; long 241.26°E. Huntoon Valley, Nev.-Calif. 1:62500 quadrangle (1958). Andesite flow in T. 4 N., R. 29 E. west of Anchorite Pass and north of Hwy 31 at 7,520′ elevation. Near top of small hill 0.95 km north of BM 7333. Reversed polarity.

[107] 5G133: lat 38.19°N.; long 241.29°E. Huntoon Valley, Nev.-Calif. 1:62500 quadrangle (1958). Basalt flow in T. 4 N., R. 29 E. southwest of Anchorite Pass. North of Hwy 31 at 7,530′ elevation and 0.45 km N21°E from BM 7401. Overlies unit 5G102. Reversed polarity. Age = 3.59 ± 0.11 Ma. Polarity and age ref. 7, unit W21.

[108] 9E224: lat 36.13°N.; long 242.24°E. Haiwee Res., Calif. 1:62500 quadrangle (1951). Andesite flow along stream near SW cor. sec. 3, T. 21 S., R. 39 E. Normal polarity. Age = 3.54 ± 0.08 Ma (ref. 19, locality 40).

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[109] 3X001: lat 35.58°N.; long 249.95°E. White Cone, Ariz. 1:62500 quadrangle (1966). Basalt flow in SW¼NE¼ sec. 12, T. 25 N., R. 21 E. approximately 0.8 km N16°E from White Cone Peak. Reversed polarity.

[110] 3X013: lat 35.58°N.; long 249.95°E. White Cone, Ariz. 1:62500 quadrangle (1966). Basalt flow overlying unit 3X001 at same locality. Reversed polarity.

[111] 3X020: lat 35.25°N; long 252.31°E. San Mateo, N. Mex. 1:24000 quadrangle (1963). Lava flow capping La Jara Mesa sampled near center sec. 21, T. 12 N., R. 8 W. Normal polarity.

[112] 3X031: lat 35.22°N; long 252.26°E. Lobo Springs, N. Mex. 1:24000 quadrangle (1957). East side of East Grants Ridge, immediately southwest of perlite quarry. NE cor. sec. 1, T. 11 N., R. 9 W. Normal polarity.

[113] 3X038: lat 35.22°N; long 252.26°E. Lobo Springs, N. Mex. 1:24000 quadrangle (1957). Perlite sampled in quarry approximately 0.6 km due west of NE cor. sec. 1, T. 11 N., R. 9 W. Normal polarity. Age = 2.81 ± 0.08 Ma. Polarity and age ref. 7, unit W12.

[114] 3X209: lat 35.87°N.; long 253.81°E. White Rock, N. Mex. 1:24000 quadrangle (1952). Basalt flow underlying the Bandelier Tuff sampled along Hwy 4 north of Los Alamos Canyon. NW¼NW¼ sec. 22, T. 19 N., R. 7 E. Reversed polarity.

[115] 3X216: lat 36.10°N.; long 253.86°E. San Juan Pueblo, N. Mex. 1:62500 quadrangle

(1953). Basalt flow in the Santa Fe Formation sampled along Hwy 84 at La Chuachia. Normal polarity.

[116] 3X245: lat 35.81°N.; long 253.76°E. White Rock, N. Mex. 1:24000 quadrangle (1952). Basalt flow underlying the Bandelier Tuff on north side of Hwy 4 SW of White Rock. 0.35 km west from BM 6456 at Potrillo Canyon. Reversed polarity.

[117] 3X252: lat 35.98°N.; long 253.66°E. Frijoles, N. Mex. 1:62500 quadrangle (1953). Quartz latite flow on north side Santa Clara Canyon and approximately 2.55 km S37°E from summit elevation 10653. Normal polarity. Age = 5.65 ± 0.82 Ma. Polarity and age ref. 7, unit W1.

[118] 3X259: lat 35.99°N.; long 253.68°E. Frijoles, N. Mex. 1:62500 quadrangle (1953). Andesite flow on north side of Santa Clara Canyon and 3.05 km S76°E from summit elevation 10653. Normal polarity. Age = 6.87 ± 0.33 Ma. Polarity and age ref. 7, unit W2.

[119] 4D082: lat 36.17°N.; long 253.65°E. Abiquiu, N. Mex. 1:24000 quadrangle (1953). Basalt flow on SE side of Polvadera Road and 0.4 km east of Abiquiu Creek at Agua Caliente. Normal polarity.

[120] 4D090: lat 36.16°N.; long 253.66°E. Abiquiu, N. Mex. 1:24000 quadrangle (1953). Basalt flow east of Polvadera Road. Sampled 120 m east of BM 7445 on west rim of Cañon Seco. Stratigraphically above unit 4D082. Normal polarity.

[121] 4D098: lat 36.16°N.; long 253.58°E. Cañones, N. Mex. 1:24000 quadrangle (1953). Basalt flow forming NE rim of Polvadera Mesa. Sampled 0.9 km N42°E from BM 7909. Normal polarity. Age = 7.63 ± 0.16 Ma. Polarity and age ref. 7, unit W3.

[122] 4D106: lat 36.10°N.; long 253.55°E. Polvadera Peak, N. Mex. 1:24000 quadrangle (1953). Quartz latite on east side of Polvadera Creek and 0.3 km S31°E from BM 7976 south of Polvadera Sawmill Camp. Normal polarity. Age = 5.16 ± 0.13 Ma. Polarity and age ref. 7, unit W4.

[123] 4D114: lat 36.04°N.; long 253.69°E. Vallecitos, N. Mex. 1:24000 quadrangle (1953). Basalt flow on NE side Rio Del Oso Road and 1.0 km N2°E from summit 8123 on Los Posos. Normal polarity.

[124] 4D122: lat 36.07°N.; long 253.57°E. Polvadera Peak, N. Mex. 1:24000 quadrangle (1953). Rhyolite dome NE of El Rechuelos and 2.1 km N72°W from VABM 11232 on Polvadera Peak at 9100′ elevation. Reversed polarity. Age = 2.08 ± 0.06 Ma. Polarity and age ref. 7, unit W5.

[125] 4D130: lat 36.11°N.; long 253.58°E. Polvadera Peak, N. Mex. 1:24000 quadrangle (1953). Rhyodacite in bottom of Canada del Ojitos and 0.7 km S40°W from summit 8770. Normal polarity. Age = 6.82 ± 0.17 Ma. Polarity and age ref. 7, unit W6.

[126] 4D140: lat 35.74°N.; long 253.55°E. Cañada, N. Mex. 1:24000 quadrangle (1953). Andesite flow SW side of road in Bland Canyon and 0.6 km N65°W from NW corner sec. 5, T. 17 N., R. 5 E. Reversed polarity.

[127] 4D148: lat 35.74°N.; long 253.55°E. Cañada, N. Mex. 1:24000 quadrangle (1953). Dacite on east side of road in Bland Canyon and 0.35 km S61°W from NW corner sec. 5, T. 17 N., R. 5 E. Reversed polarity. Age = 9.11 ± 0.19 Ma. Polarity and age ref. 7, unit W7.

[128] 4D155: lat 35.70°N.; long 253.60°E. Cañada, N. Mex. 1:24000 quadrangle (1953). Rhyodacite near Rancho Cañada in Cochiti Canyon. On north side of road and 0.5 km N15°E from elevation 6563 on Horn Mesa. Normal polarity. Age = 9.33 ± 0.19 Ma. Polarity and age ref. 7, unit W8.

[129] 4D165: lat 35.76°N.; long 253.63°E. Frijoles, N. Mex. 1:62500 quadrangle (1953). Andesite sampled in roadcut immediately north of lookout tower on St. Peters Dome. Normal polarity. Age = 8.69 ± 0.38 Ma. Polarity and age ref. 7, unit W9.

[130] 4D174: lat 35.83°N.; long 253.60°E. Frijoles, N. Mex. 1:62500 quadrangle (1953). Rhyodacite of Cerro Grande sampled on NW side of Hwy 4. East of Sandoval-Los Alamos County line and 0.2 km NE from BM 8812. Normal polarity. Age = 3.77 ± 0.12 Ma. Polarity and age ref. 7, unit W10.

[131] 4D183: lat 35.76°N.; long 253.74°E. Frijoles, N. Mex. 1:62500 quadrangle (1953). Basalt flow sampled at Upper Frijoles Falls in Cañon de los Frijoles. Normal polarity.

[132] 4D191: lat 35.77°N.; long 253.74°E. Frijoles, N. Mex. 1:62500 quadrangle (1953). Basalt flow sampled on NE side of Cañon de los Frijoles and 170 m S77°E from BM 5877 at 6000′ elevation. Normal polarity.

[133] 4D218: lat 36.16°N.; long 253.59°E. Cañones, N. Mex. 1:24000 quadrangle (1953). Andesite flow forming NE rim of canyon of Polvadera Creek. Sampled at 7950′ elevation and 1.85 km

N45°E from BM 7909 on Polvadera Mesa. Normal polarity.

[134] 4D227: lat 36.10°N.; long 253.61°E. Polvadera Peak, N. Mex. 1:24000 quadrangle (1953). Rhyodacite sampled on SW flank of hill 8454 east of Abiquiu Creek at 8220′ elevation. 1.5 km N87°E from BM 8945 on El Lagunito Palo Quemador. Normal polarity. Age = 5.78 ± 0.18 Ma. Polarity and age ref. 7, unit W11.

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